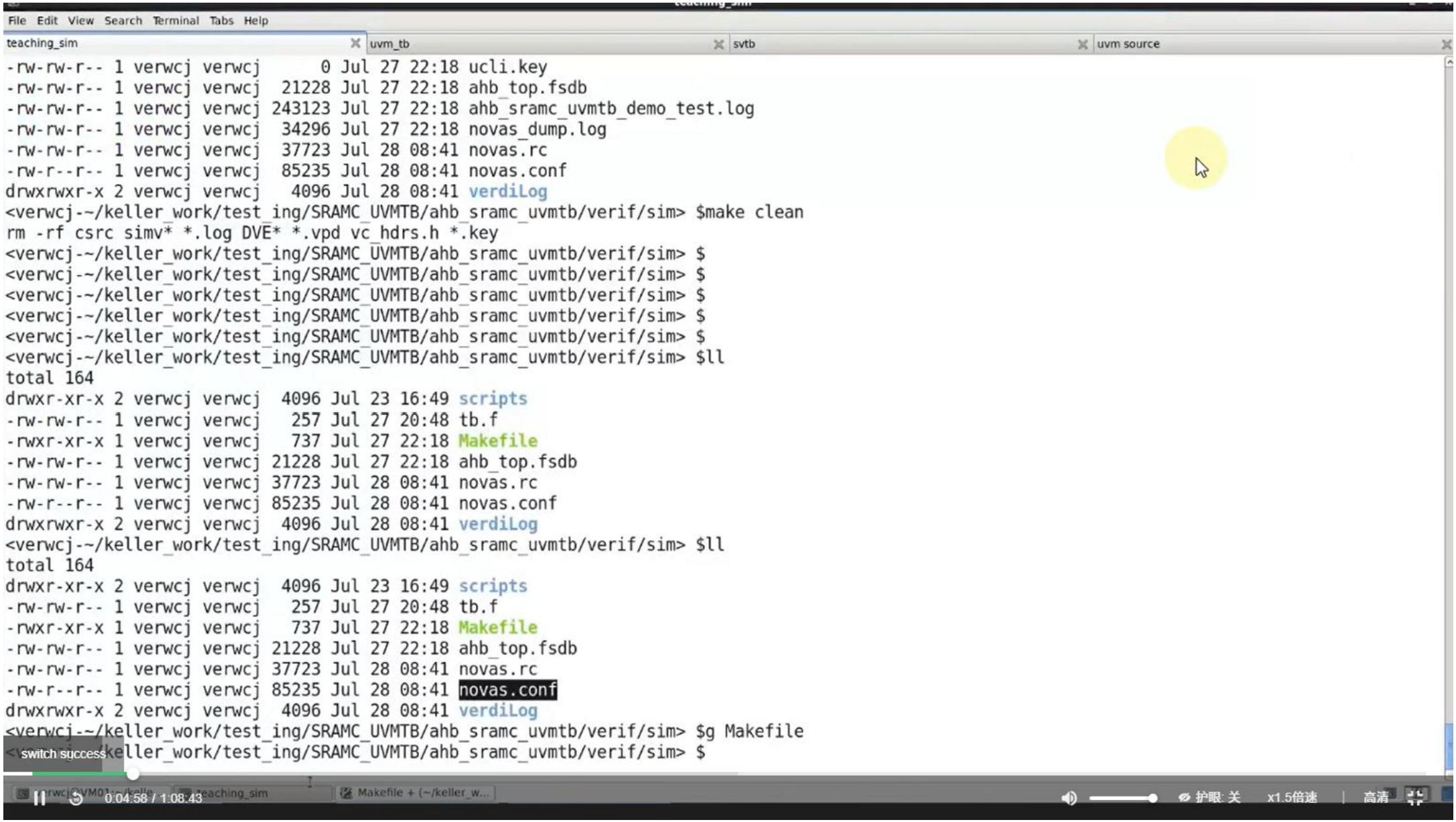
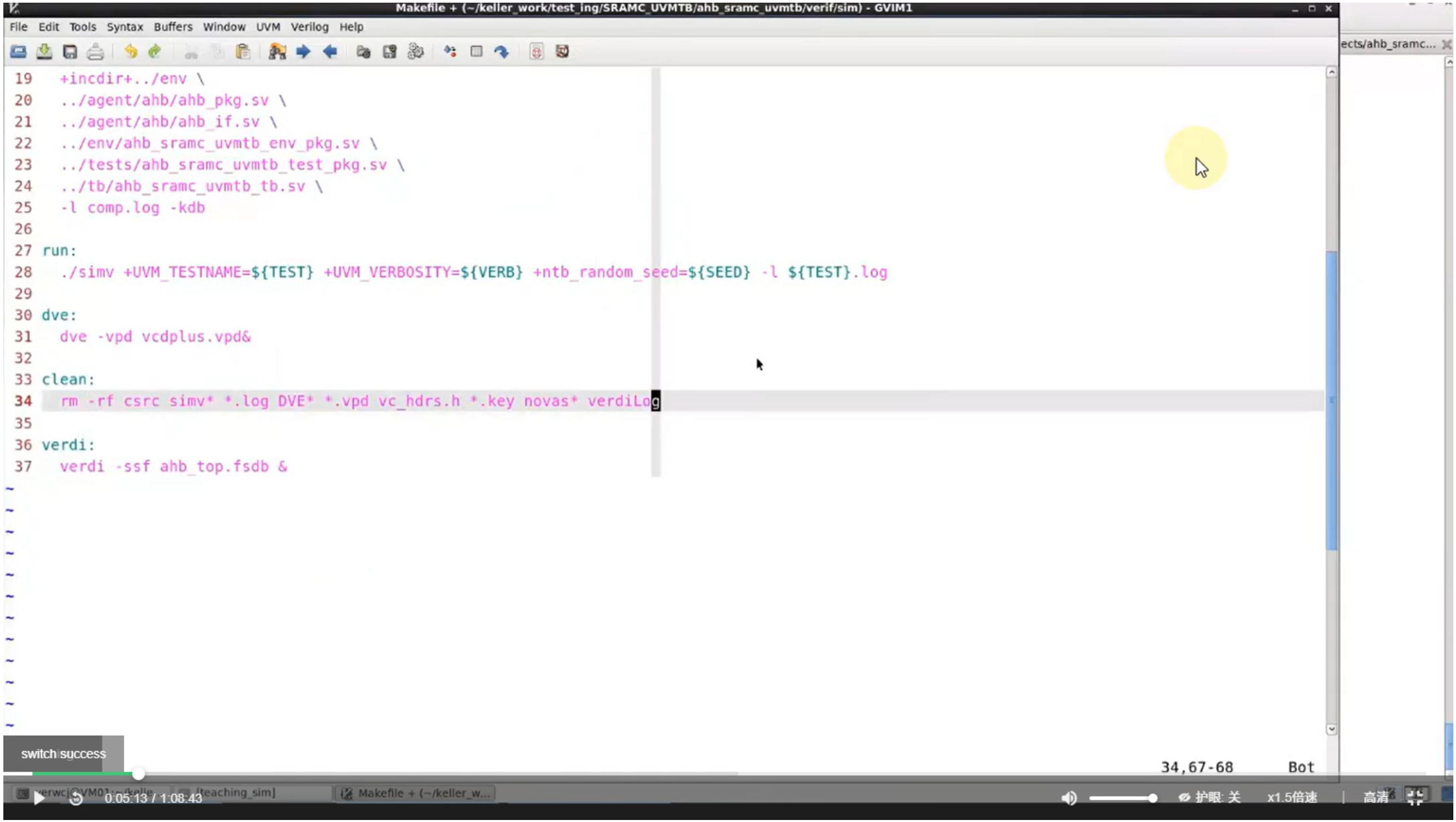
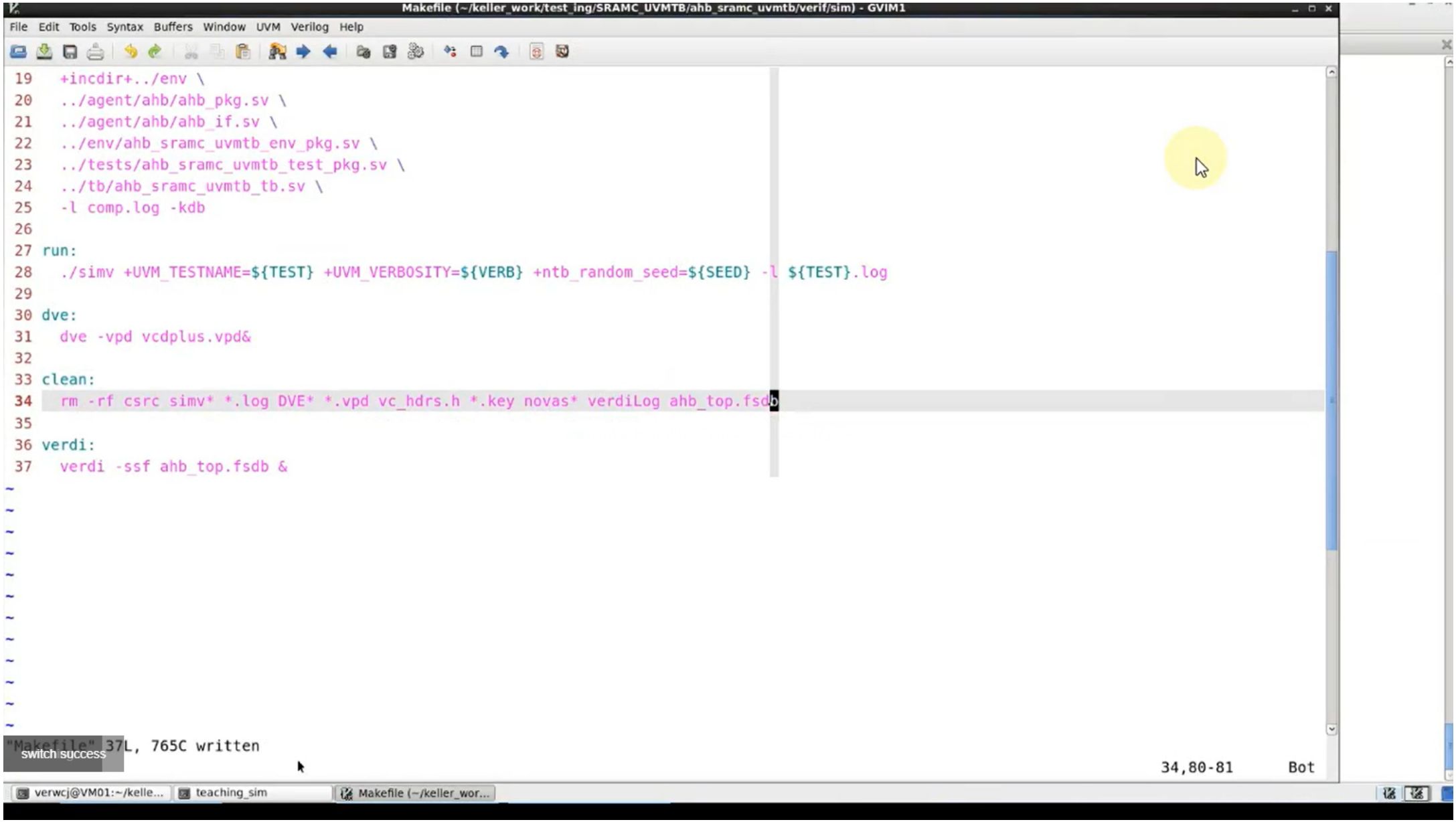
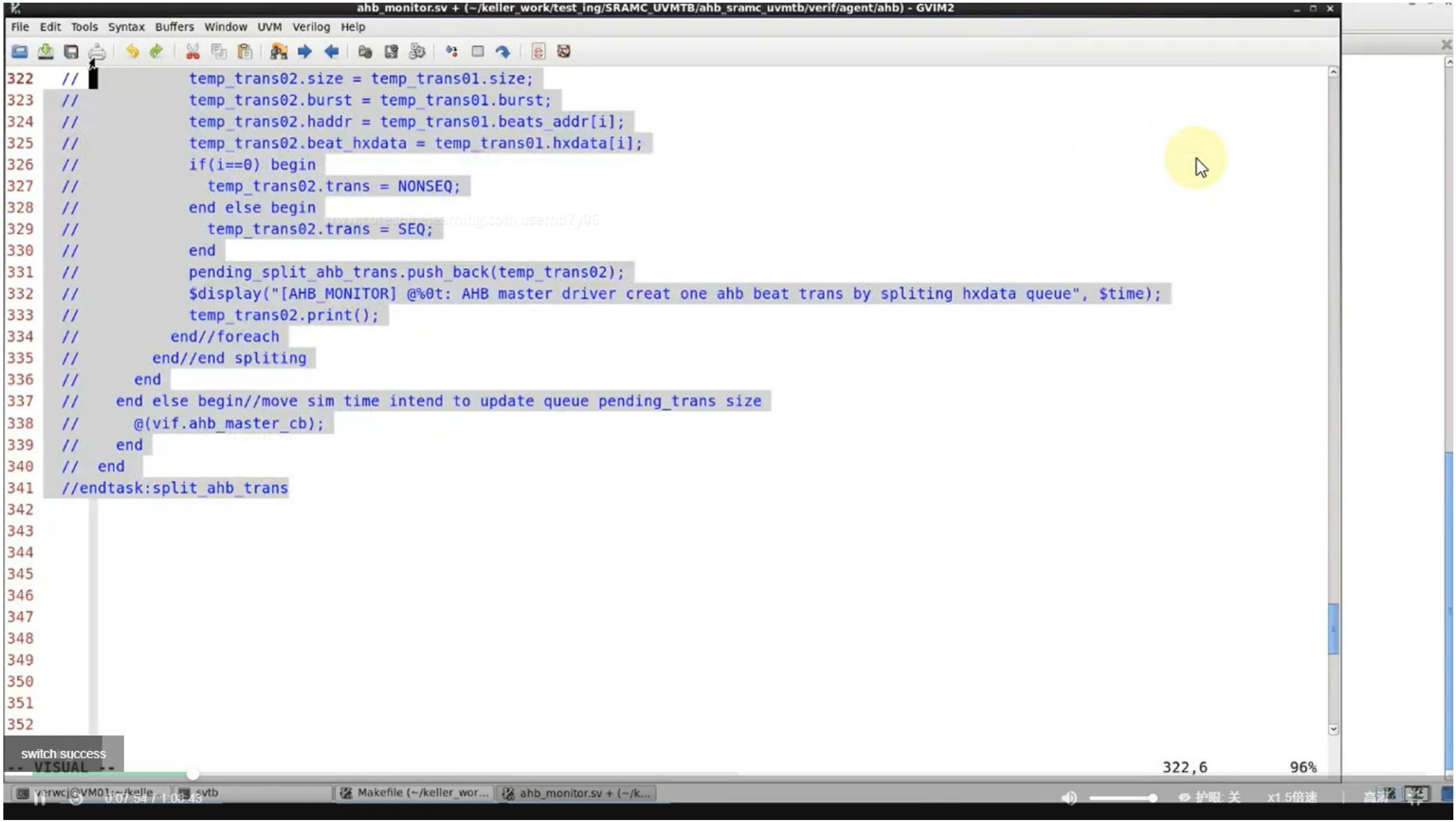
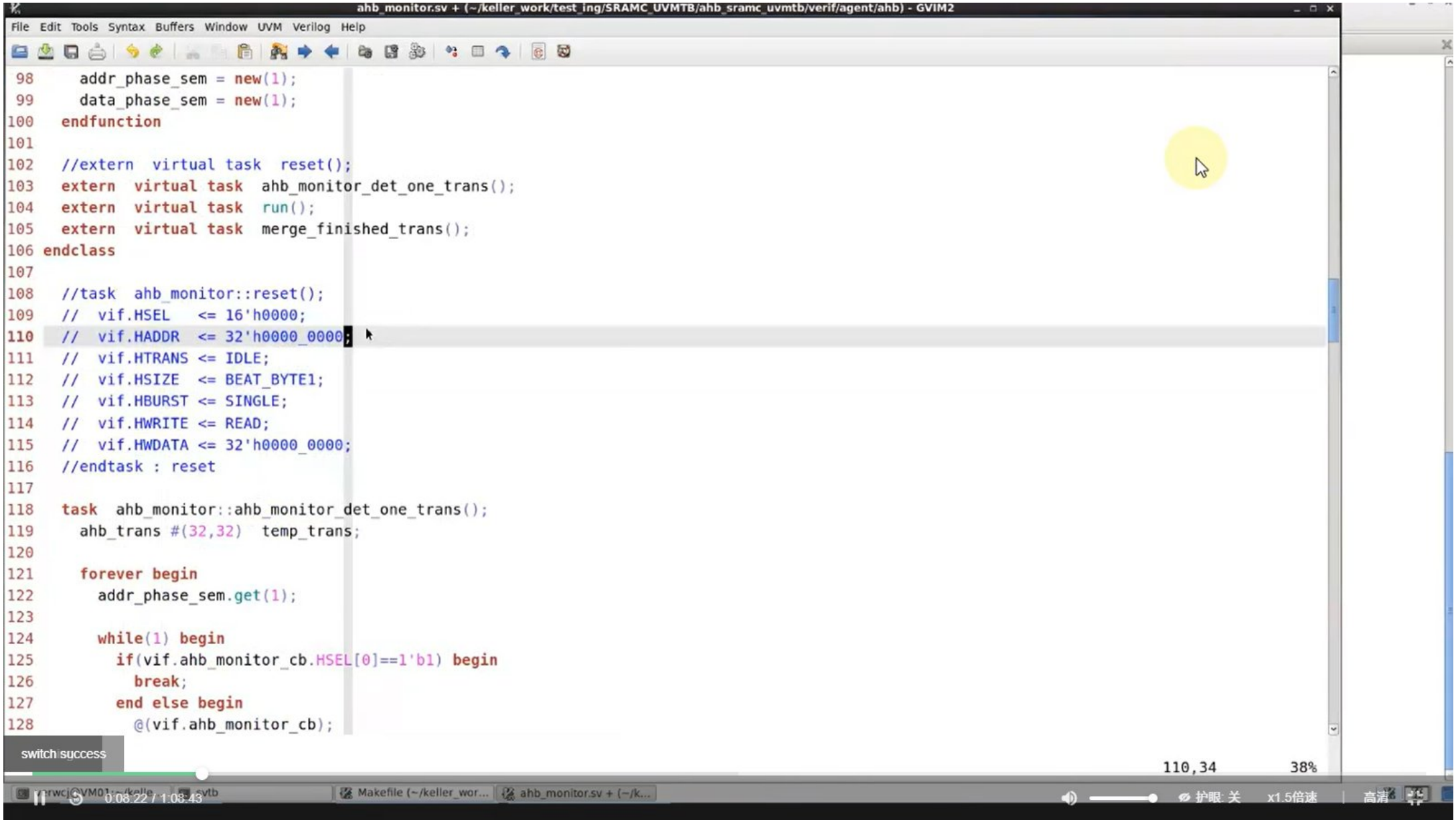
本节核心内容：
1. 改造monitor代码
2. 根据波形进行debug

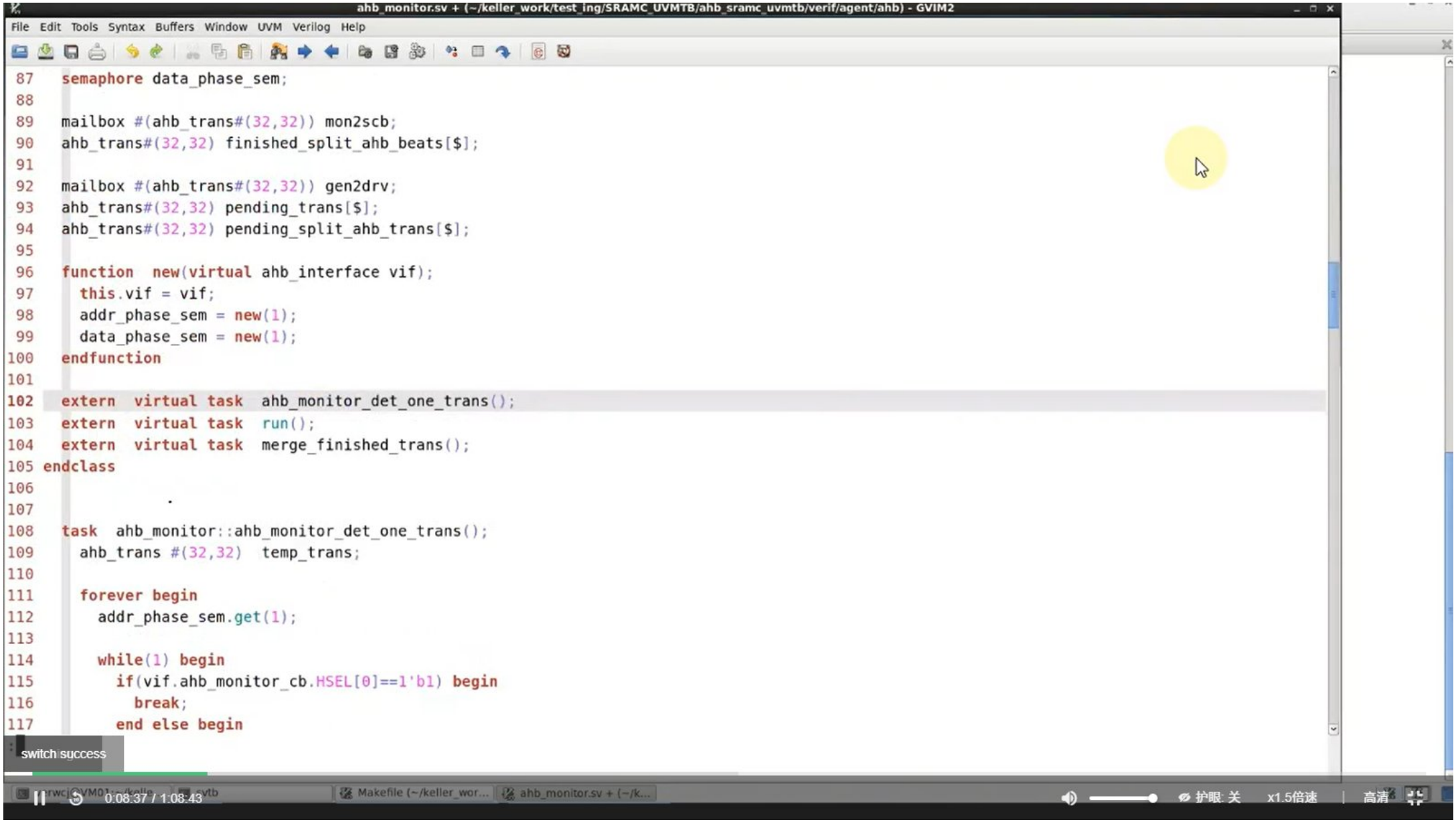


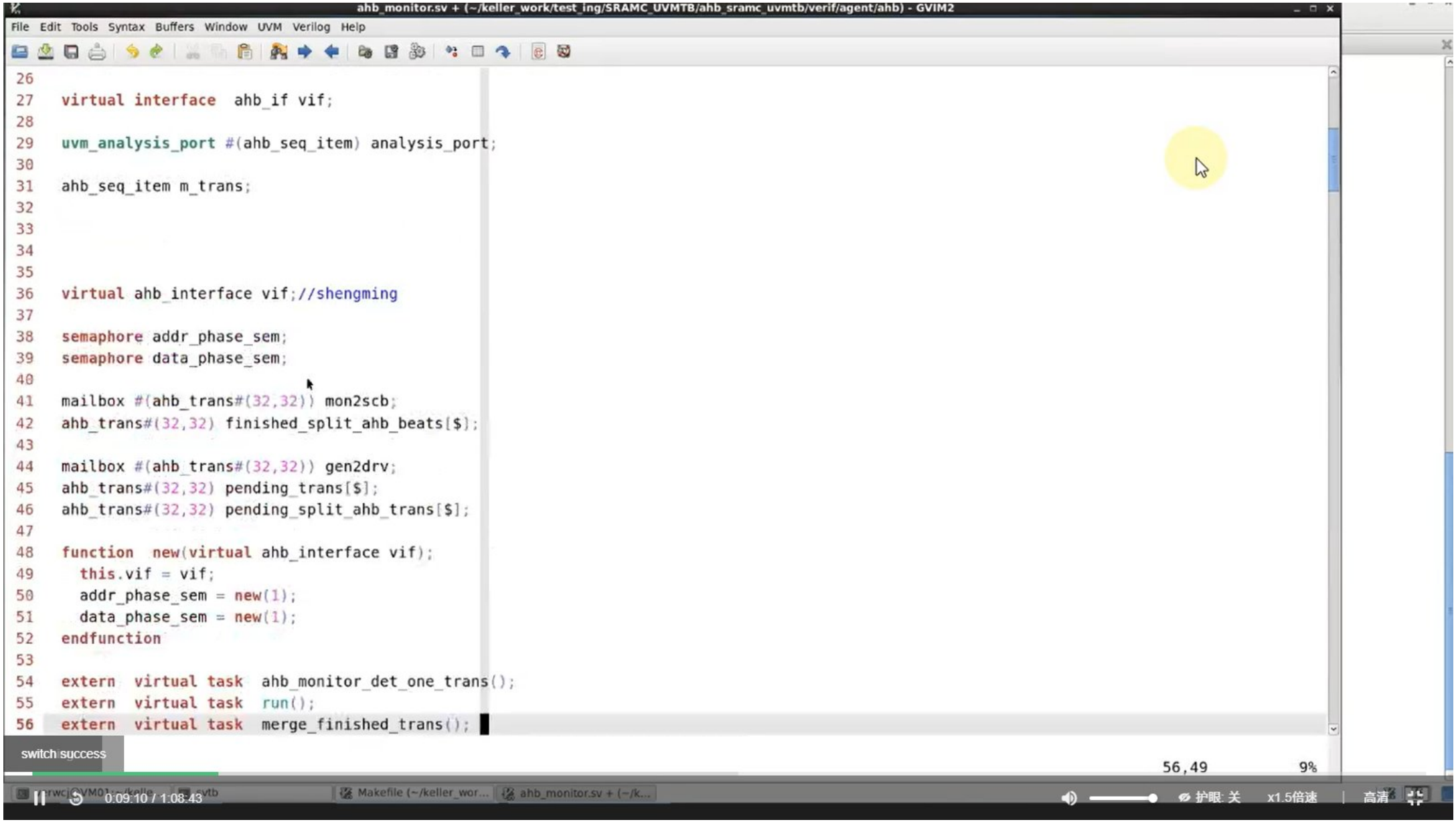

Makefile + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim) - GVIM1
File Edit Tools Syntax Buffers Window UVM Verilog Help
+incdir+../env \
../agent/ahb/ahb_pkg.sv \
../agent/ahb/ahb_if.sv \
../env/ahb_sramc_uvmtb_env_pkg.sv \
../tests/ahb_sramc_uvmtb_test_pkg.sv \
../tb/ahb_sramc_uvmtb_tb.sv \
-l comp.log -kdb
run:
./simv +UVM_TESTNAME=${TEST} +UVM_VERBOSITY=${VERB} +ntb_random_seed=${SEED} -l ${TEST}.log
dve:
dve -vpd vcdplus.vpd&
clean:
rm -rf csrc simv* *.log DVE* *.vpd vc_hdrs.h *.key novas* verdiLog
verdi:
verdi -ssf ahb_top.fsdb &
switch success
34,67-68 Bot
0:05:13 / 1:08:43
护眼: 关
x1.5倍速
高清

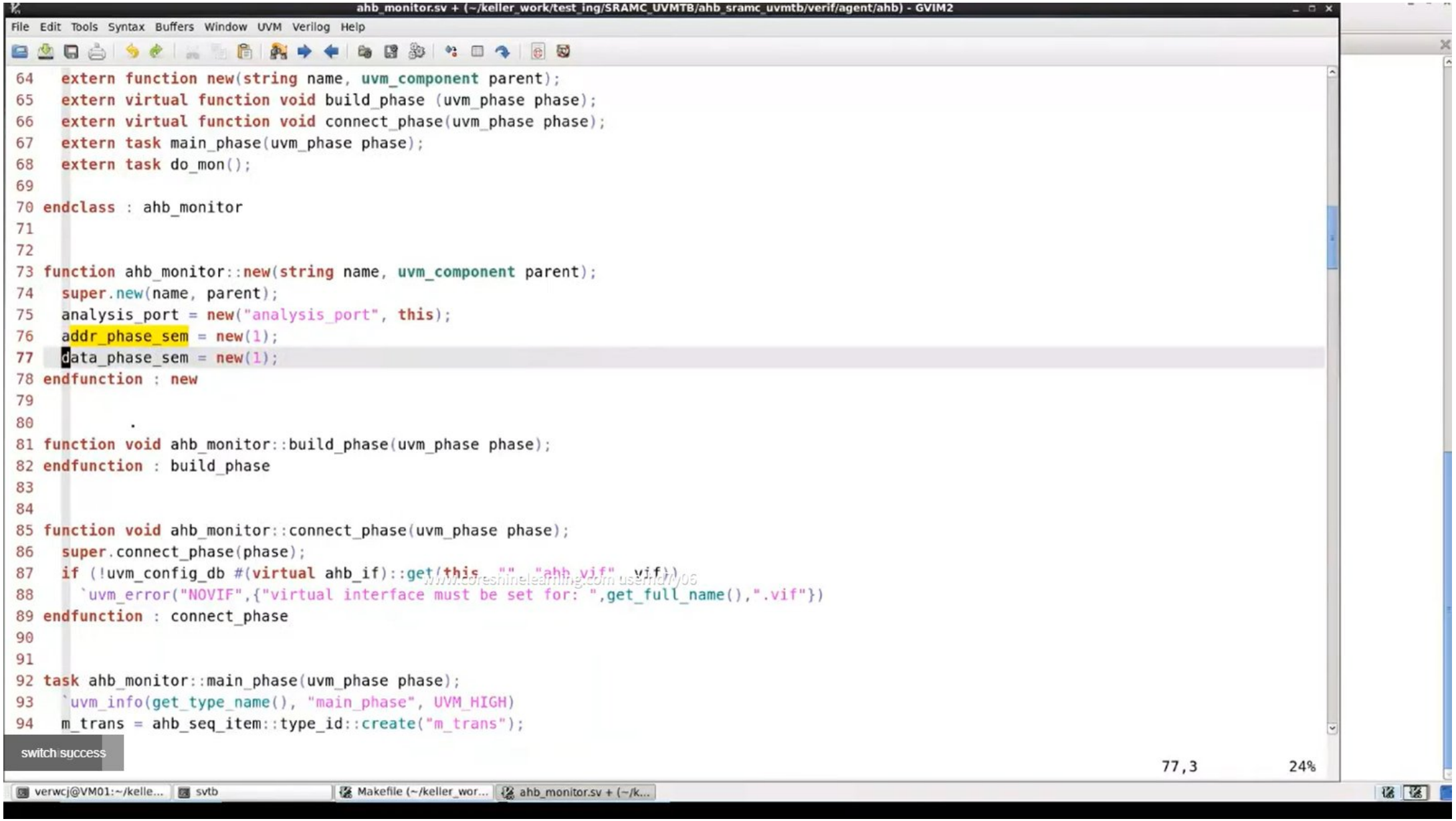

Makefile (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim) - GVIM1
File Edit Tools Syntax Buffers Window UVM Verilog Help
19 +incdir+../env \
20 ../agent/ahb/ahb_pkg.sv \
21 ../agent/ahb/ahb_if.sv \
22 ../env/ahb_sramc_uvmtb_env_pkg.sv \
23 ../tests/ahb_sramc_uvmtb_test_pkg.sv \
24 ../tb/ahb_sramc_uvmtb_tb.sv \
25 -l comp.log -kdb
26
27 run:
28 ./simv +UVM_TESTNAME=${TEST} +UVM_VERBOSITY=${VERB} +ntb_random_seed=${SEED} -l ${TEST}.log
29
30 dve:
31 dve -vpd vcdplus.vpd&
32
33 clean:
34 rm -rf csrc simv* *.log DVE* *.vpd vc_hdrs.h *.key novas* verdiLog ahb_top.fsdb
35
36 verdi:
37 verdi -ssf ahb_top.fsdb &
"Makefile" 37L, 765C written
switch success
34,80-81 Bot
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...

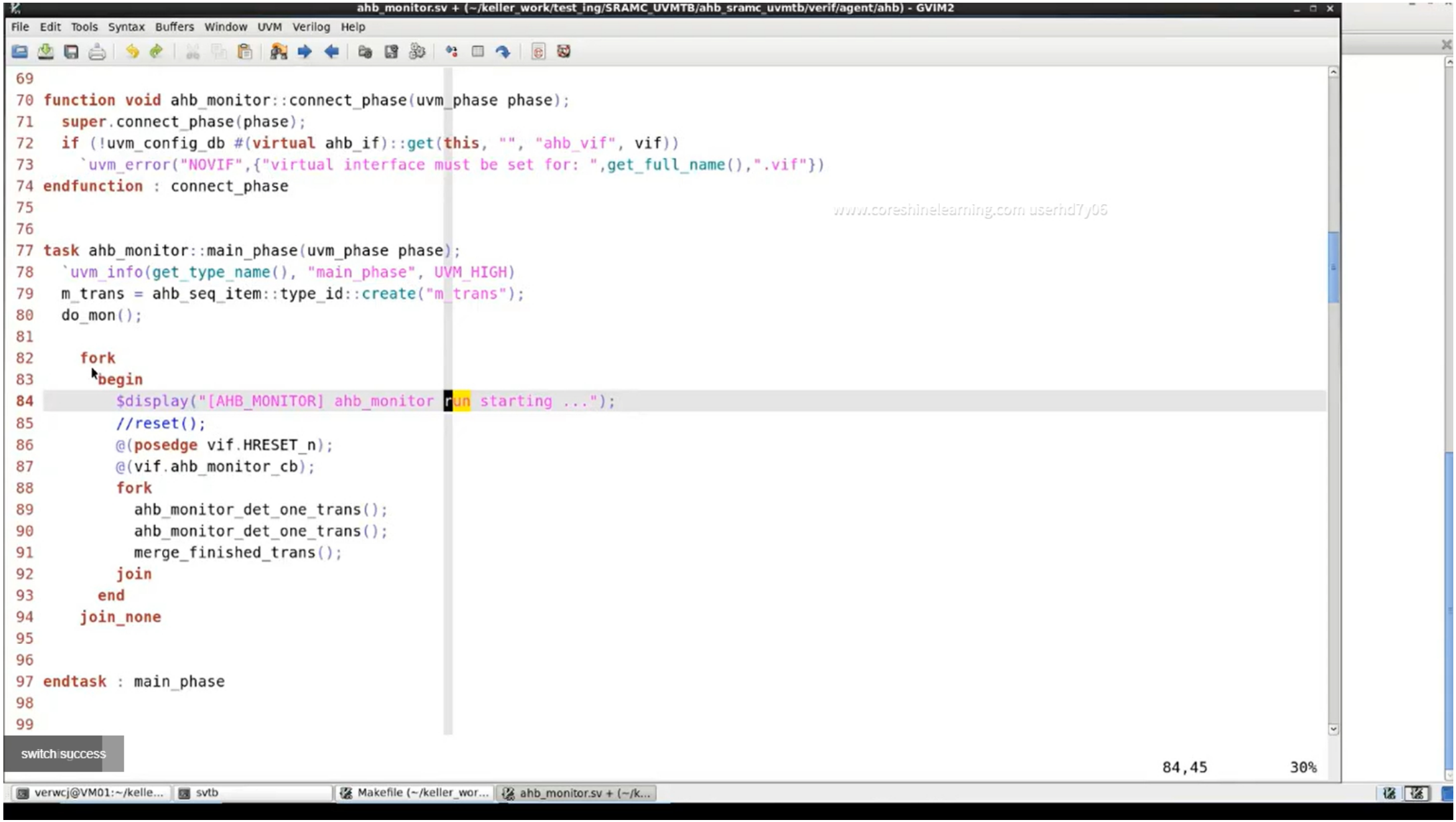

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
322 // temp_trans02.size = temp_trans01.size;
323 // temp_trans02.burst = temp_trans01.burst;
324 // temp_trans02.haddr = temp_trans01.beats_addr[i];
325 // temp_trans02.beat_hxdata = temp_trans01.hxdata[i];
326 // if(i==0) begin
327 // temp_trans02.trans = NONSEQ;
328 // end else begin
329 // temp_trans02.trans = SEQ;
330 // end
331 // pending_split_ahb_trans.push_back(temp_trans02);
332 // $display("[AHB_MONITOR] @%0t: AHB master driver creat one ahb beat trans by spliting hxdata queue", $time);
333 // temp_trans02.print();
334 // end//foreach
335 // end//end spliting
336 // end
337 // end else begin//move sim time intend to update queue pending_trans size
338 // @(vif.ahb_master_cb);
339 // end
340 // end
341 //endtask:split_ahb_trans
342
343
344
345
346
347
348
349
350
351
352
switch success
-- VISUAL --
322,6 96%
svtb
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

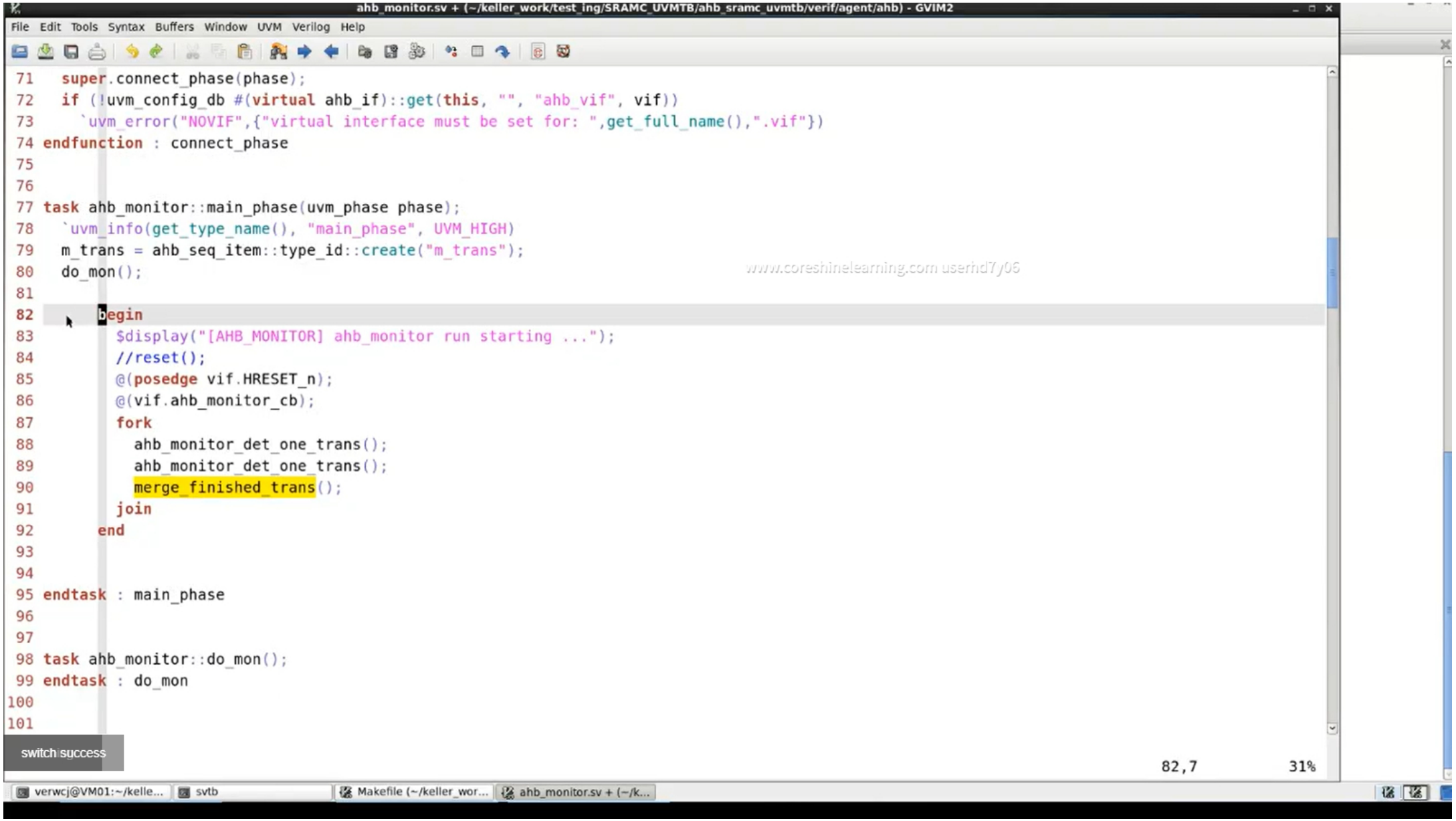

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
addr_phase_sem = new(1);
data_phase_sem = new(1);
endfunction
//extern virtual task reset();
extern virtual task ahb_monitor_det_one_trans();
extern virtual task run();
extern virtual task merge_finished_trans();
endclass
//task ahb_monitor::reset();
// vif.HSEL <= 16'h0000;
// vif.HADDR <= 32'h0000_0000;
// vif.HTRANS <= IDLE;
// vif.HSIZE <= BEAT_BYTE1;
// vif.HBURST <= SINGLE;
// vif.HWRITE <= READ;
// vif.HWDATA <= 32'h0000_0000;
//endtask : reset
task ahb_monitor::ahb_monitor_det_one_trans();
ahb_trans #(32,32) temp_trans;
forever begin
addr_phase_sem.get(1);
while(1) begin
if(vif.ahb_monitor_cb.HSEL[0]==1'b1) begin
break;
end else begin
@(vif.ahb_monitor_cb);
switch success
110,34 38%
0:08:22 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

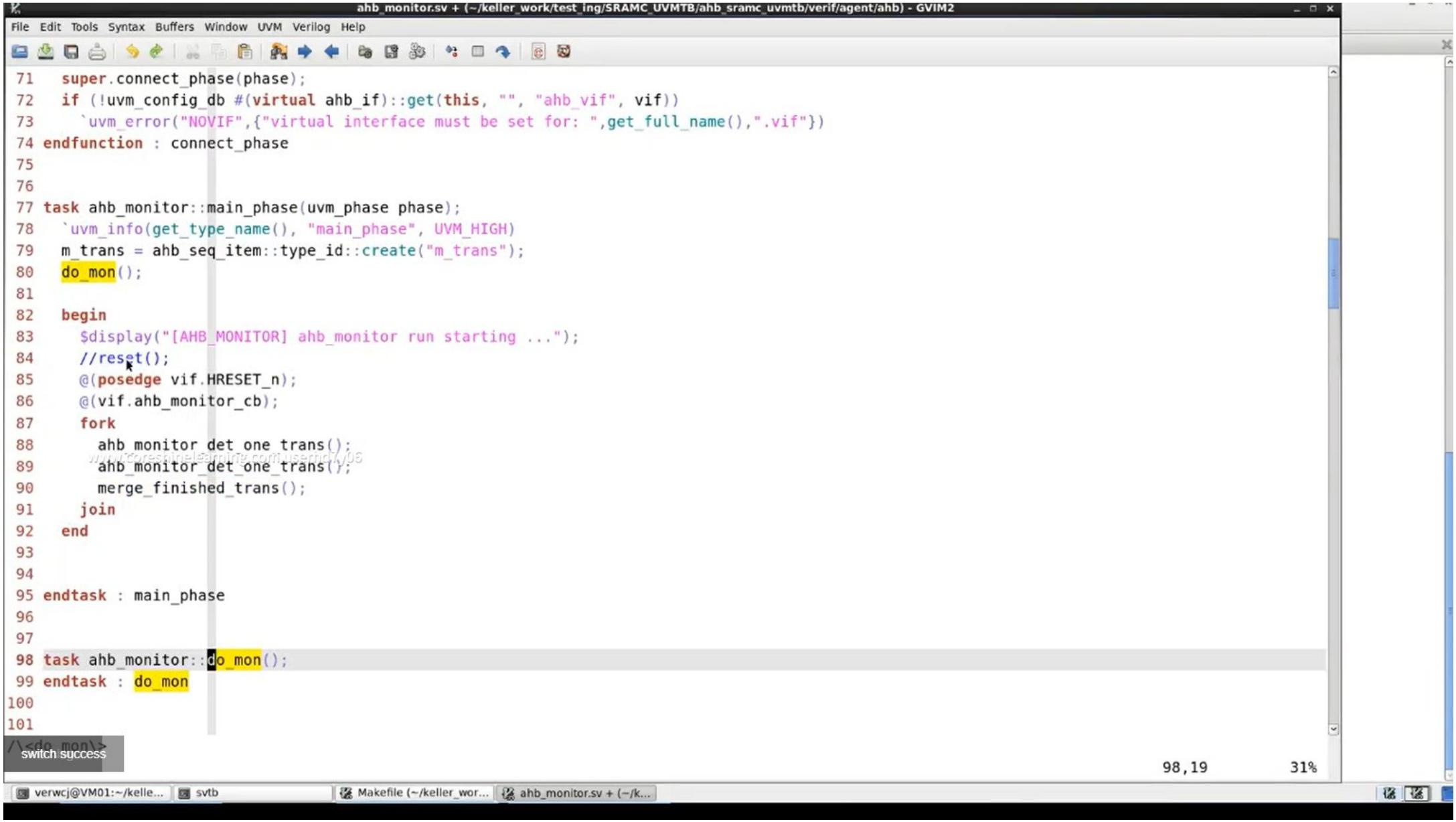

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
87 semaphore data_phase_sem;
88
89 mailbox #(ahb_trans#(32,32)) mon2scb;
90 ahb_trans#(32,32) finished_split_ahb_beats[$];
91
92 mailbox #(ahb_trans#(32,32)) gen2drv;
93 ahb_trans#(32,32) pending_trans[$];
94 ahb_trans#(32,32) pending_split_ahb_trans[$];
95
96 function new(virtual ahb_interface vif);
97 this.vif = vif;
98 addr_phase_sem = new(1);
99 data_phase_sem = new(1);
100 endfunction
101
102 extern virtual task ahb_monitor_det_one_trans();
103 extern virtual task run();
104 extern virtual task merge_finished_trans();
105 endclass
106
107
108 task ahb_monitor::ahb_monitor_det_one_trans();
109 ahb_trans #(32,32) temp_trans;
110
111 forever begin
112 addr_phase_sem.get(1);
113
114 while(1) begin
115 if(vif.ahb_monitor_cb.HSEL[0]==1'b1) begin
116 break;
117 end else begin
switch success
0:08:37 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

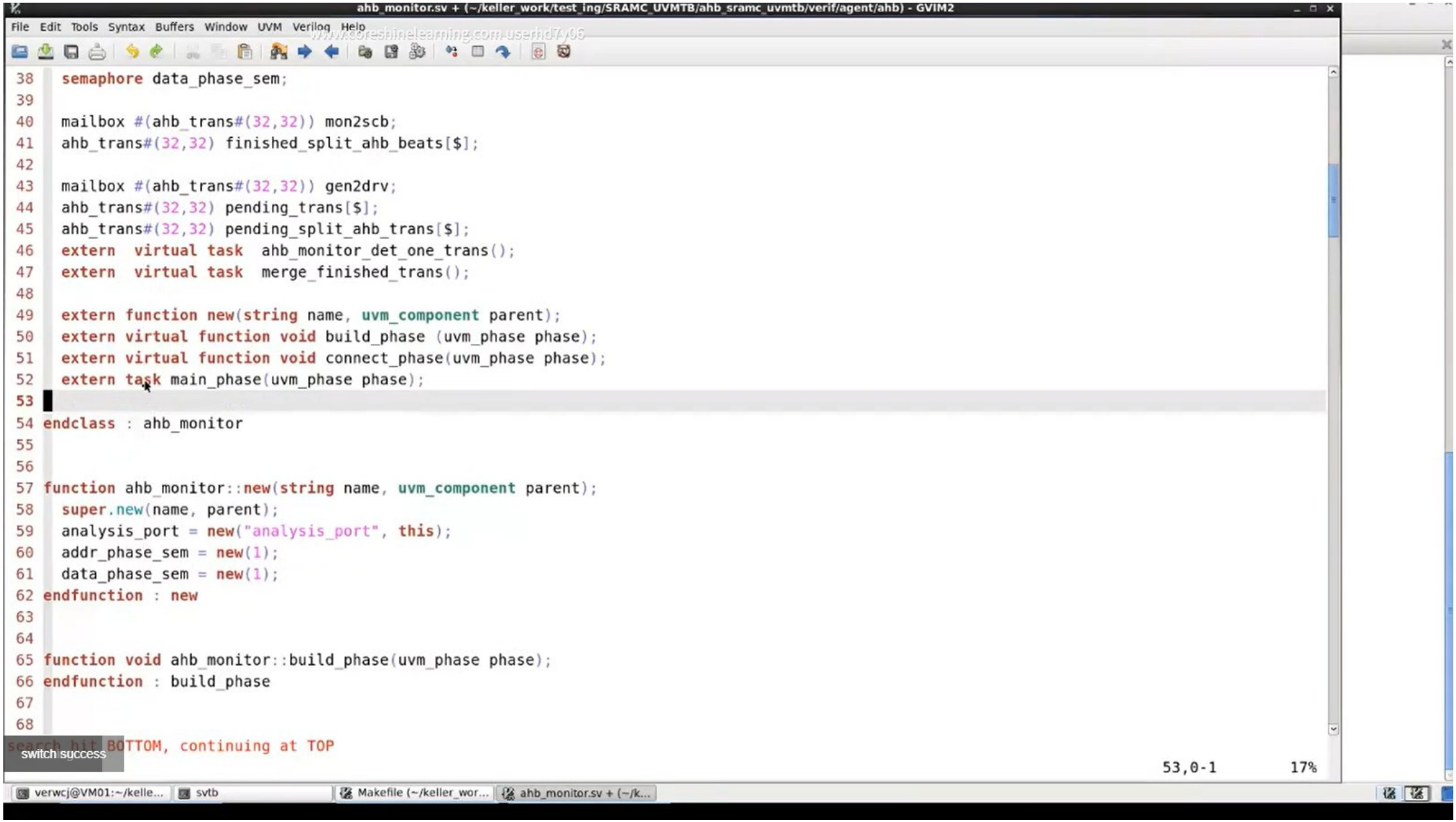

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
26
27 virtual interface ahb_if vif;
28
29 uvm_analysis_port #(ahb_seq_item) analysis_port;
30
31 ahb_seq_item m_trans;
32
33
34
35
36 virtual ahb_interface vif;//shengming
37
38 semaphore addr_phase_sem;
39 semaphore data_phase_sem;
40
41 mailbox #(ahb_trans#(32,32)) mon2scb;
42 ahb_trans#(32,32) finished_split_ahb_beats[$];
43
44 mailbox #(ahb_trans#(32,32)) gen2drv;
45 ahb_trans#(32,32) pending_trans[$];
46 ahb_trans#(32,32) pending_split_ahb_trans[$];
47
48 function new(virtual ahb_interface vif);
49 this.vif = vif;
50 addr_phase_sem = new(1);
51 data_phase_sem = new(1);
52 endfunction
53
54 extern virtual task ahb_monitor_det_one_trans();
55 extern virtual task run();
56 extern virtual task merge_finished_trans();
switch success
56,49 9%
0:09:10 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

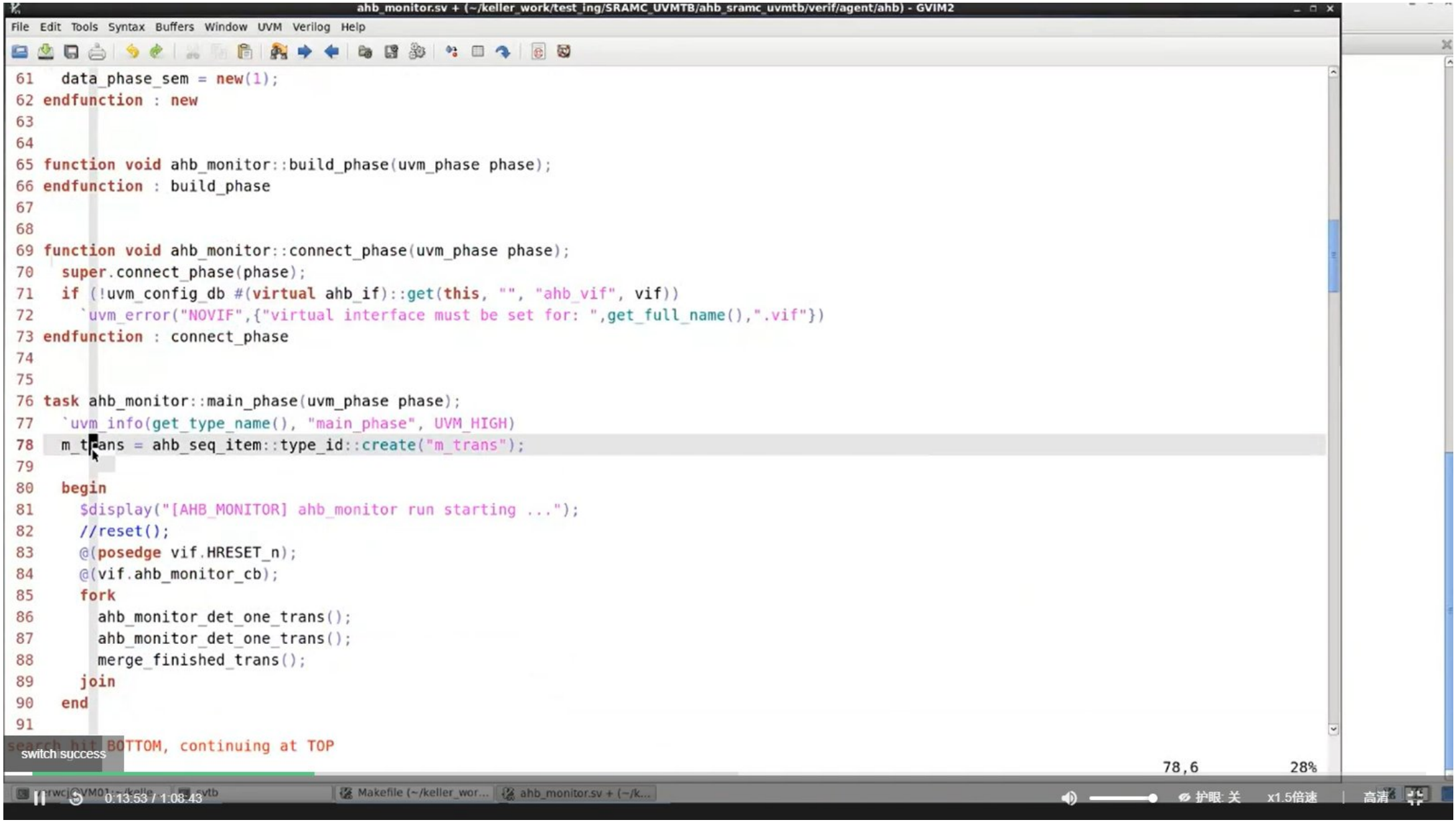

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
64 extern function new(string name, uvm_component parent);
65 extern virtual function void build_phase (uvm_phase phase);
66 extern virtual function void connect_phase(uvm_phase phase);
67 extern task main_phase(uvm_phase phase);
68 extern task do_mon();
69
70 endclass : ahb_monitor
71
72
73 function ahb_monitor::new(string name, uvm_component parent);
74 super.new(name, parent);
75 analysis_port = new("analysis_port", this);
76 addr_phase_sem = new(1);
77 data_phase_sem = new(1);
78 endfunction : new
79
80 .
81 function void ahb_monitor::build_phase(uvm_phase phase);
82 endfunction : build_phase
83
84
85 function void ahb_monitor::connect_phase(uvm_phase phase);
86 super.connect_phase(phase);
87 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
88 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
89 endfunction : connect_phase
90
91
92 task ahb_monitor::main_phase(uvm_phase phase);
93 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
94 m_trans = ahb_seq_item::type_id::create("m_trans");
switch success
77,3 24%
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

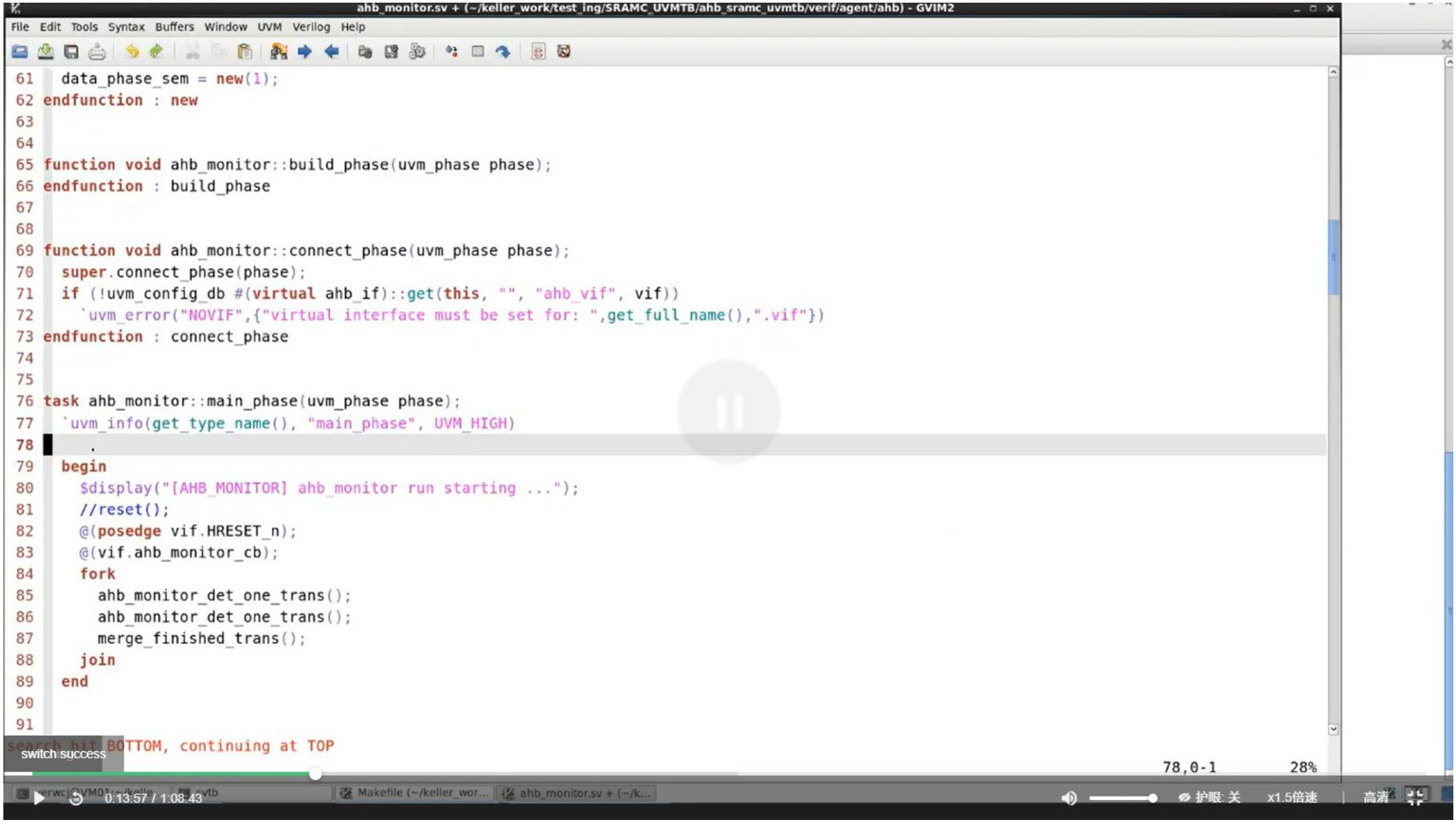

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
69
70 function void ahb_monitor::connect_phase(uvm_phase phase);
71 super.connect_phase(phase);
72 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
73 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
74 endfunction : connect_phase
75
76
77 task ahb_monitor::main_phase(uvm_phase phase);
78 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
79 m_trans = ahb_seq_item::type_id::create("m_trans");
80 do_mon();
81
82 fork
83 begin
84 $display("[AHB_MONITOR] ahb_monitor run starting ...");
85 //reset();
86 @(posedge vif.HRESET_n);
87 @(vif.ahb_monitor_cb);
88 fork
89 ahb_monitor_det_one_trans();
90 ahb_monitor_det_one_trans();
91 merge_finished_trans();
92 join
93 end
94 join_none
95
96
97 endtask : main_phase
98
99
www.coreshinelearning.com userhd7j06
switch success
84,45 30%
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

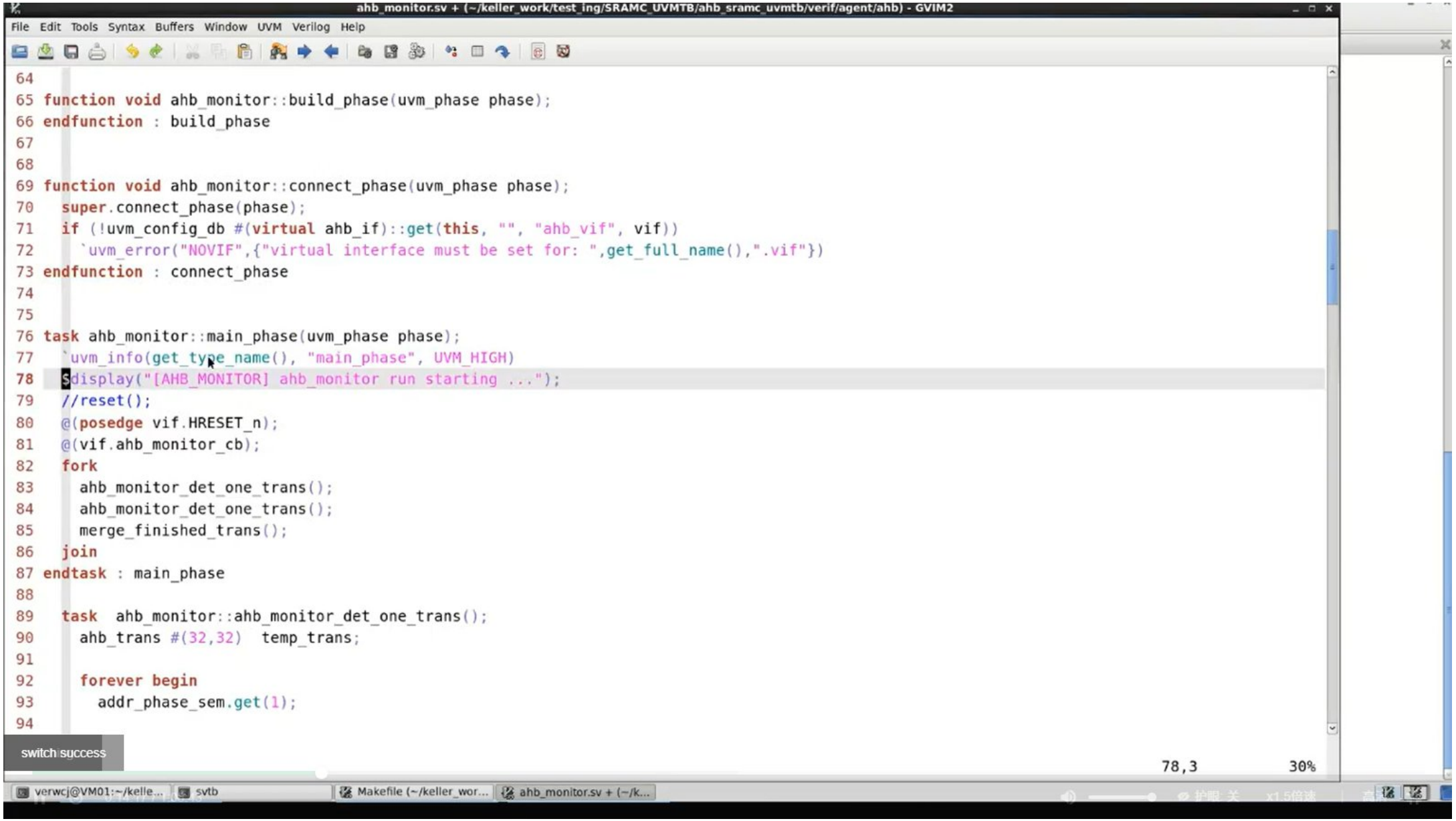

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
71 super.connect_phase(phase);
72 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
73 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
74 endfunction : connect_phase
75
76
77 task ahb_monitor::main_phase(uvm_phase phase);
78 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
79 m_trans = ahb_seq_item::type_id::create("m_trans");
80 do_mon();
81
82 begin
83 $display("[AHB_MONITOR] ahb_monitor run starting ...");
84 //reset();
85 @(posedge vif.HRESET_n);
86 @(vif.ahb_monitor_cb);
87 fork
88 ahb_monitor_det_one_trans();
89 ahb_monitor_det_one_trans();
90 merge_finished_trans();
91 join
92 end
93
94
95 endtask : main_phase
96
97
98 task ahb_monitor::do_mon();
99 endtask : do_mon
100
101
www.coreshinelearning.com userhd7y06
switch success
82,7 31%
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

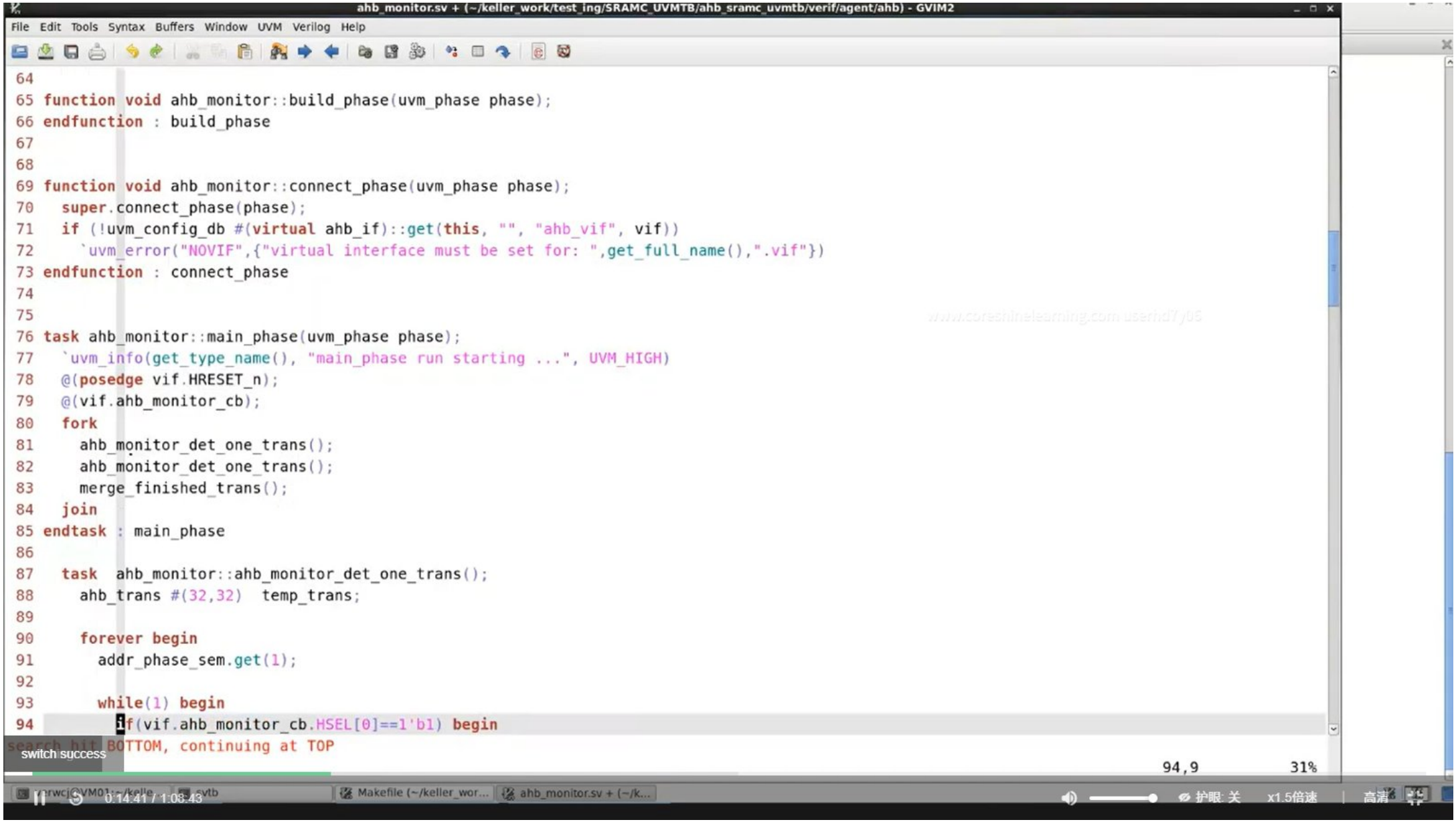

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
71 super.connect_phase(phase);
72 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
73 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
74 endfunction : connect_phase
75
76
77 task ahb_monitor::main_phase(uvm_phase phase);
78 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
79 m_trans = ahb_seq_item::type_id::create("m_trans");
80 do_mon();
81
82 begin
83 $display("[AHB_MONITOR] ahb_monitor run starting ...");
84 //reset();
85 @(posedge vif.HRESET_n);
86 @(vif.ahb_monitor_cb);
87 fork
88 ahb_monitor_det_one_trans();
89 ahb_monitor_det_one_trans();
90 merge_finished_trans();
91 join
92 end
93
94
95 endtask : main_phase
96
97
98 task ahb_monitor::do_mon();
99 endtask : do_mon
100
101
/\<do_mon\>
switch success
98,19 31%
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

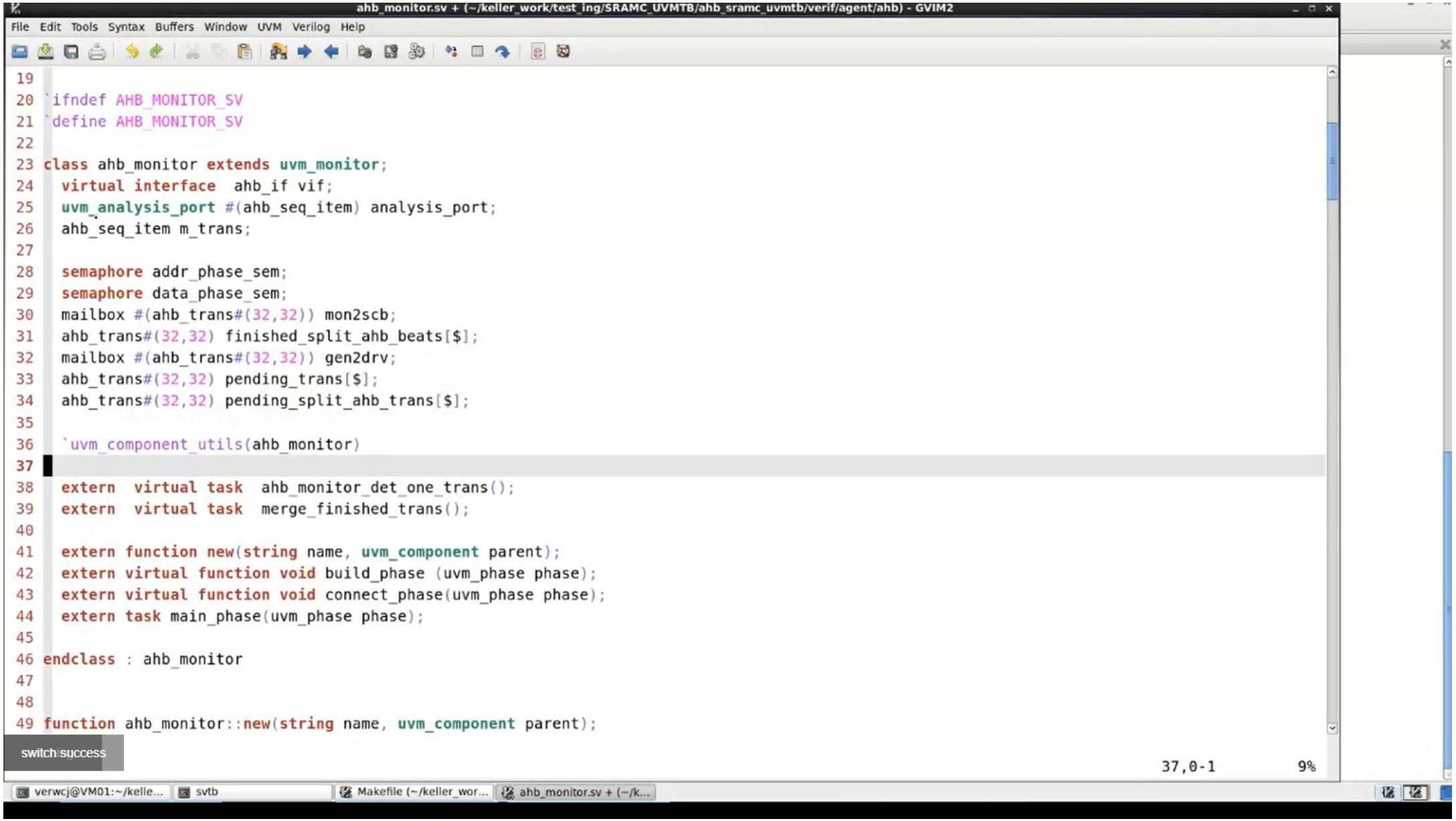

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
www.coreshinelearning.com userid7y06
38 semaphore data_phase_sem;
39
40 mailbox #(ahb_trans#(32,32)) mon2scb;
41 ahb_trans#(32,32) finished_split_ahb_beats[$];
42
43 mailbox #(ahb_trans#(32,32)) gen2drv;
44 ahb_trans#(32,32) pending_trans[$];
45 ahb_trans#(32,32) pending_split_ahb_trans[$];
46 extern virtual task ahb_monitor_det_one_trans();
47 extern virtual task merge_finished_trans();
48
49 extern function new(string name, uvm_component parent);
50 extern virtual function void build_phase (uvm_phase phase);
51 extern virtual function void connect_phase(uvm_phase phase);
52 extern task main_phase(uvm_phase phase);
53
54 endclass : ahb_monitor
55
56
57 function ahb_monitor::new(string name, uvm_component parent);
58 super.new(name, parent);
59 analysis_port = new("analysis_port", this);
60 addr_phase_sem = new(1);
61 data_phase_sem = new(1);
62 endfunction : new
63
64
65 function void ahb_monitor::build_phase(uvm_phase phase);
66 endfunction : build_phase
67
68
search hit BOTTOM, continuing at TOP
53,0-1 17%
switch success
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

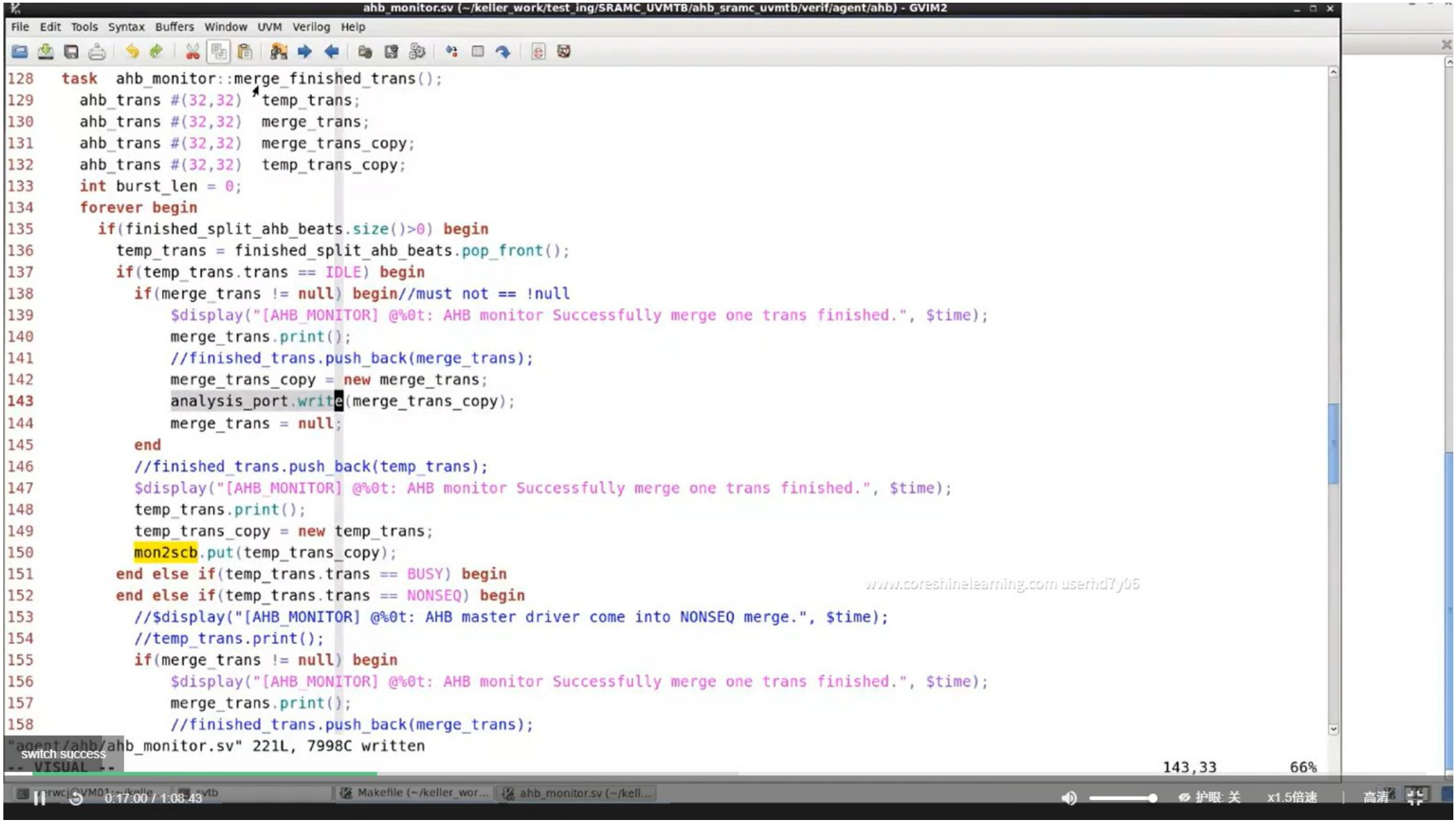

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
61 data_phase_sem = new(1);
62 endfunction : new
63
64
65 function void ahb_monitor::build_phase(uvm_phase phase);
66 endfunction : build_phase
67
68
69 function void ahb_monitor::connect_phase(uvm_phase phase);
70 super.connect_phase(phase);
71 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
72 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
73 endfunction : connect_phase
74
75
76 task ahb_monitor::main_phase(uvm_phase phase);
77 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
78 m_trans = ahb_seq_item::type_id::create("m_trans");
79
80 begin
81 $display("[AHB_MONITOR] ahb_monitor run starting ...");
82 //reset();
83 @(posedge vif.HRESET_n);
84 @(vif.ahb_monitor_cb);
85 fork
86 ahb_monitor_det_one_trans();
87 ahb_monitor_det_one_trans();
88 merge_finished_trans();
89 join
90 end
91
search hit BOTTOM, continuing at TOP
switch success
78,6 28%
0:13:53 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

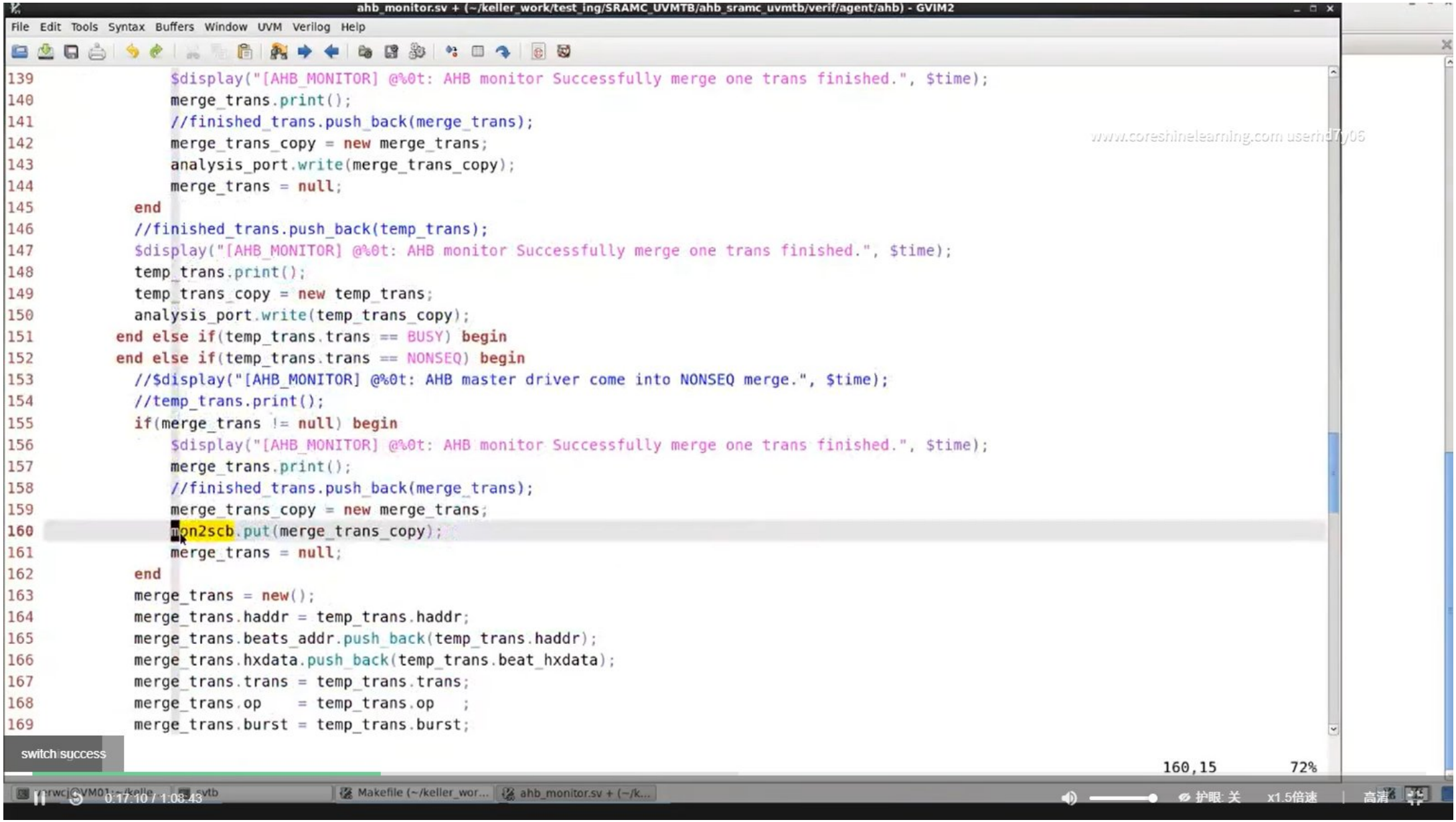

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
61 data_phase_sem = new(1);
62 endfunction : new
63
64
65 function void ahb_monitor::build_phase(uvm_phase phase);
66 endfunction : build_phase
67
68
69 function void ahb_monitor::connect_phase(uvm_phase phase);
70 super.connect_phase(phase);
71 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
72 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
73 endfunction : connect_phase
74
75
76 task ahb_monitor::main_phase(uvm_phase phase);
77 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
78 .
79 begin
80 $display("[AHB_MONITOR] ahb_monitor run starting ...");
81 //reset();
82 @(posedge vif.HRESET_n);
83 @(vif.ahb_monitor_cb);
84 fork
85 ahb_monitor_det_one_trans();
86 ahb_monitor_det_one_trans();
87 merge_finished_trans();
88 join
89 end
90
91
search hit BOTTOM, continuing at TOP
switch success
78,0-1 28%
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
0:13:57 / 1:08:43
护眼: 关
x1.5倍速
高清

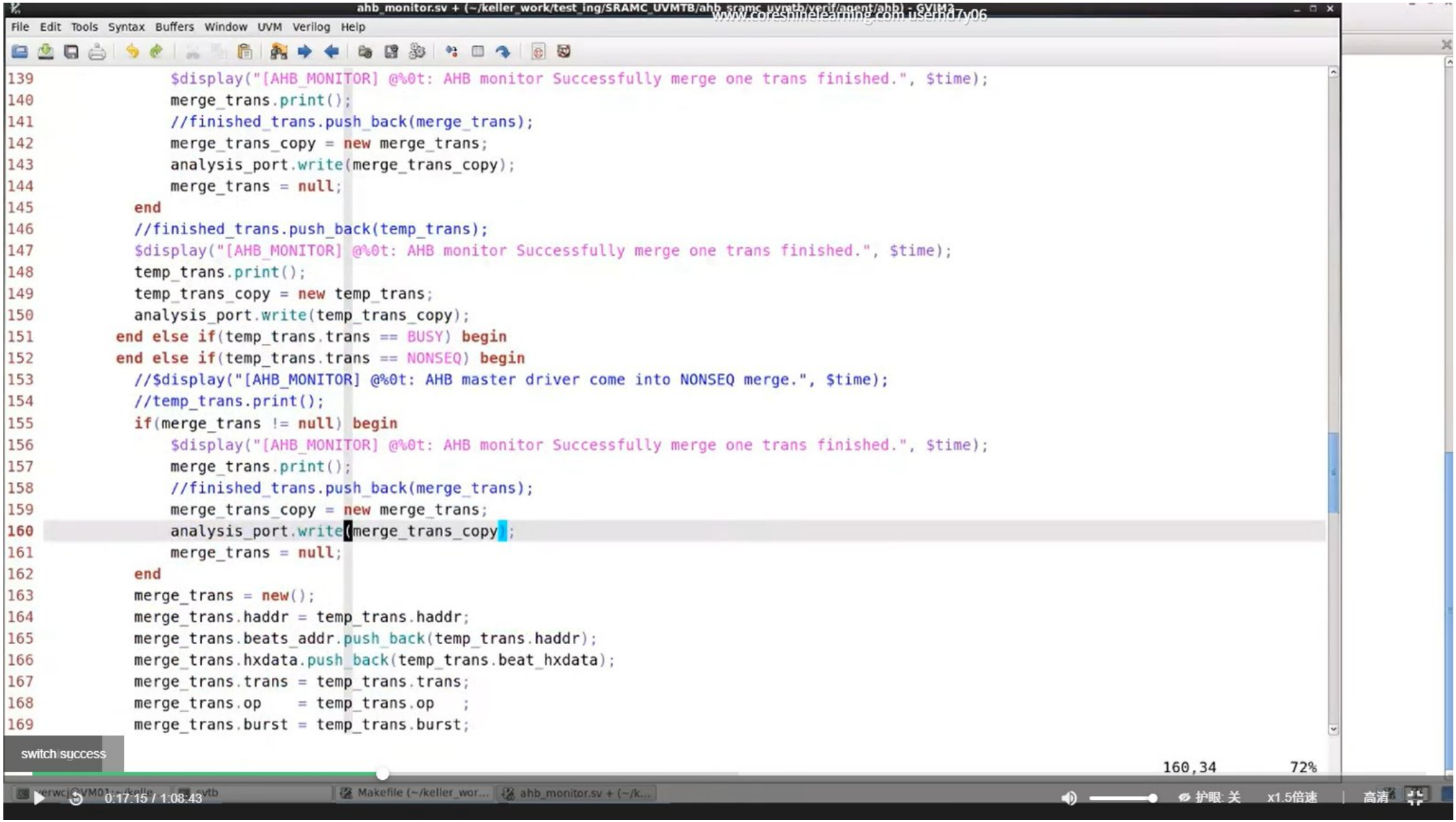

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
64
65 function void ahb_monitor::build_phase(uvm_phase phase);
66 endfunction : build_phase
67
68
69 function void ahb_monitor::connect_phase(uvm_phase phase);
70 super.connect_phase(phase);
71 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
72 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
73 endfunction : connect_phase
74
75
76 task ahb_monitor::main_phase(uvm_phase phase);
77 `uvm_info(get_type_name(), "main_phase", UVM_HIGH)
78 $display("[AHB_MONITOR] ahb_monitor run starting ...");
79 //reset();
80 @(posedge vif.HRESET_n);
81 @(vif.ahb_monitor_cb);
82 fork
83 ahb_monitor_det_one_trans();
84 ahb_monitor_det_one_trans();
85 merge_finished_trans();
86 join
87 endtask : main_phase
88
89 task ahb_monitor::ahb_monitor_det_one_trans();
90 ahb_trans #(32,32) temp_trans;
91
92 forever begin
93 addr_phase_sem.get(1);
94
switch success
78,3 30%
verwcj@VM01:~/kelle... svtb Makefile (~/keller_wor... ahb_monitor.sv + (~/k...

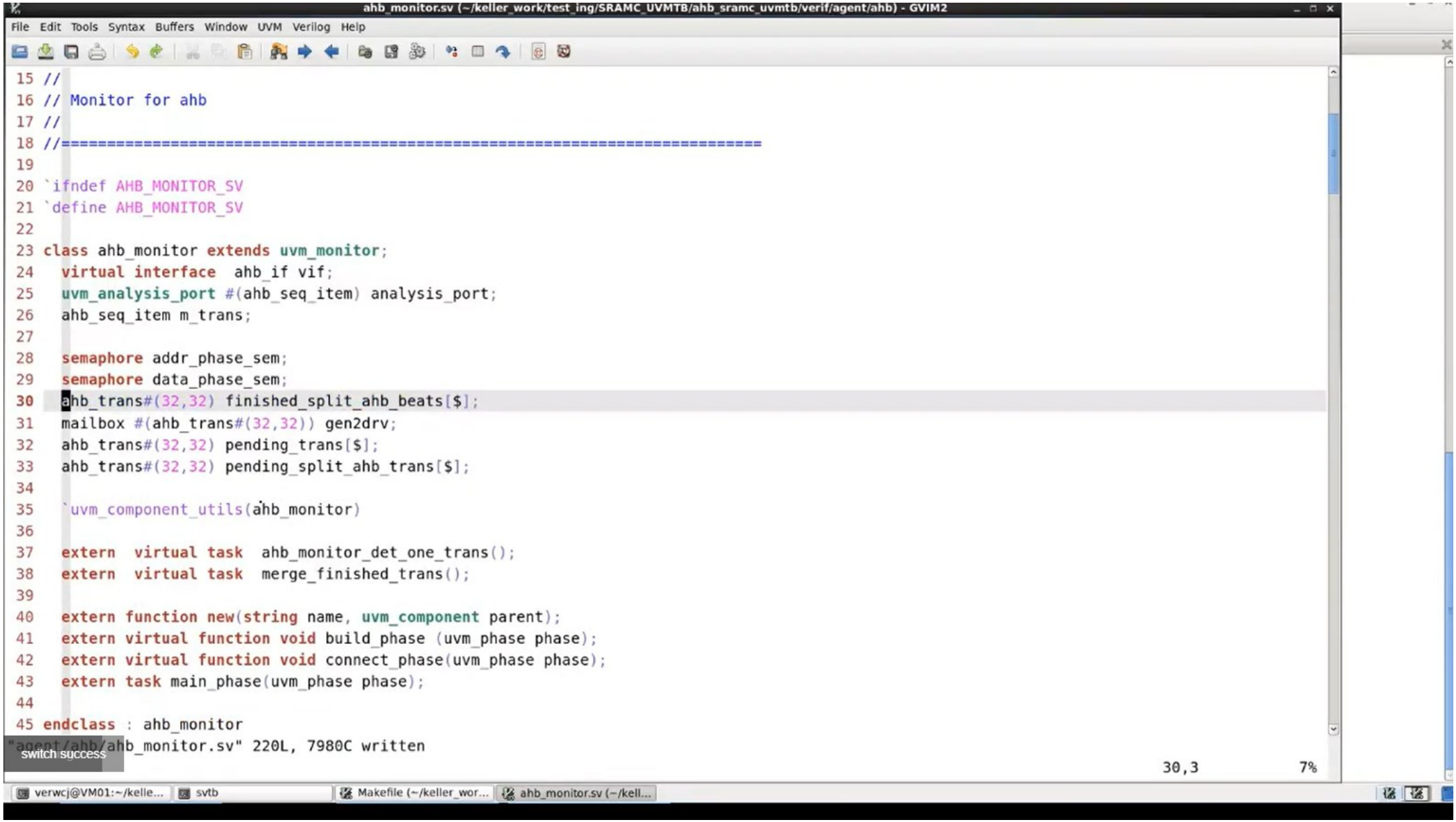

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
64
65 function void ahb_monitor::build_phase(uvm_phase phase);
66 endfunction : build_phase
67
68
69 function void ahb_monitor::connect_phase(uvm_phase phase);
70 super.connect_phase(phase);
71 if (!uvm_config_db #(virtual ahb_if)::get(this, "", "ahb_vif", vif))
72 `uvm_error("NOVIF",{"virtual interface must be set for: ",get_full_name(),".vif"})
73 endfunction : connect_phase
74
75
76 task ahb_monitor::main_phase(uvm_phase phase);
77 `uvm_info(get_type_name(), "main_phase run starting ...", UVM_HIGH)
78 @(posedge vif.HRESET_n);
79 @(vif.ahb_monitor_cb);
80 fork
81 ahb_monitor_det_one_trans();
82 ahb_monitor_det_one_trans();
83 merge_finished_trans();
84 join
85 endtask : main_phase
86
87 task ahb_monitor::ahb_monitor_det_one_trans();
88 ahb_trans #(32,32) temp_trans;
89
90 forever begin
91 addr_phase_sem.get(1);
92
93 while(1) begin
94 if(vif.ahb_monitor_cb.HSEL[0]==1'b1) begin
search hit BOTTOM, continuing at TOP
switch success
94,9 31%
0:14:41 / 1:08:43
x1.5倍速
高清

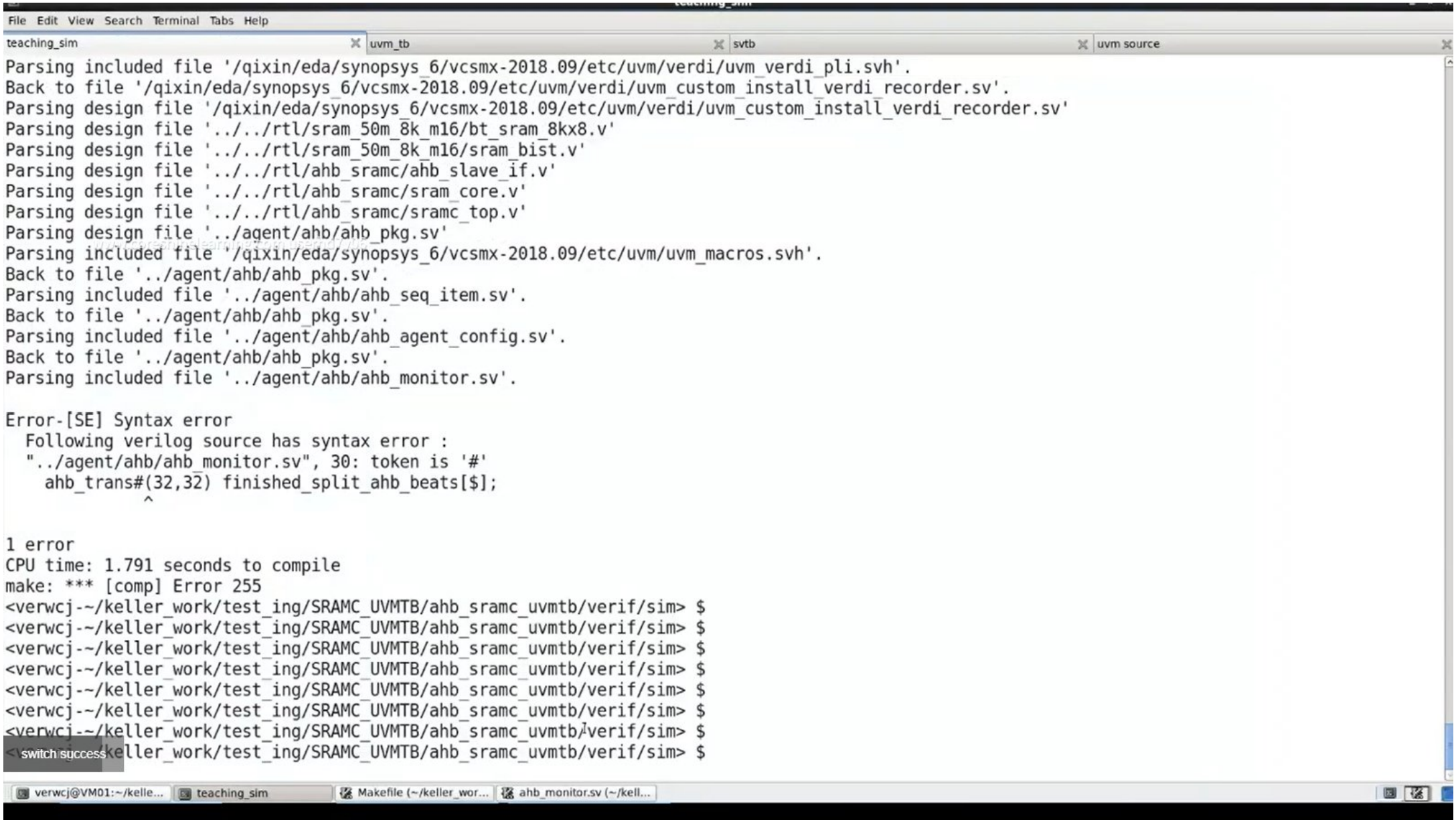

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
19
20 `ifndef AHB_MONITOR_SV
21 `define AHB_MONITOR_SV
22
23 class ahb_monitor extends uvm_monitor;
24 virtual interface ahb_if vif;
25 uvm_analysis_port #(ahb_seq_item) analysis_port;
26 ahb_seq_item m_trans;
27
28 semaphore addr_phase_sem;
29 semaphore data_phase_sem;
30 mailbox #(ahb_trans#(32,32)) mon2scb;
31 ahb_trans#(32,32) finished_split_ahb_beats[$];
32 mailbox #(ahb_trans#(32,32)) gen2drv;
33 ahb_trans#(32,32) pending_trans[$];
34 ahb_trans#(32,32) pending_split_ahb_trans[$];
35
36 `uvm_component_utils(ahb_monitor)
37
38 extern virtual task ahb_monitor_det_one_trans();
39 extern virtual task merge_finished_trans();
40
41 extern function new(string name, uvm_component parent);
42 extern virtual function void build_phase (uvm_phase phase);
43 extern virtual function void connect_phase(uvm_phase phase);
44 extern task main_phase(uvm_phase phase);
45
46 endclass : ahb_monitor
47
48
49 function ahb_monitor::new(string name, uvm_component parent);
switch success
37,0-1 9%
verwcj@VM01:~/kelle...
svtb
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...

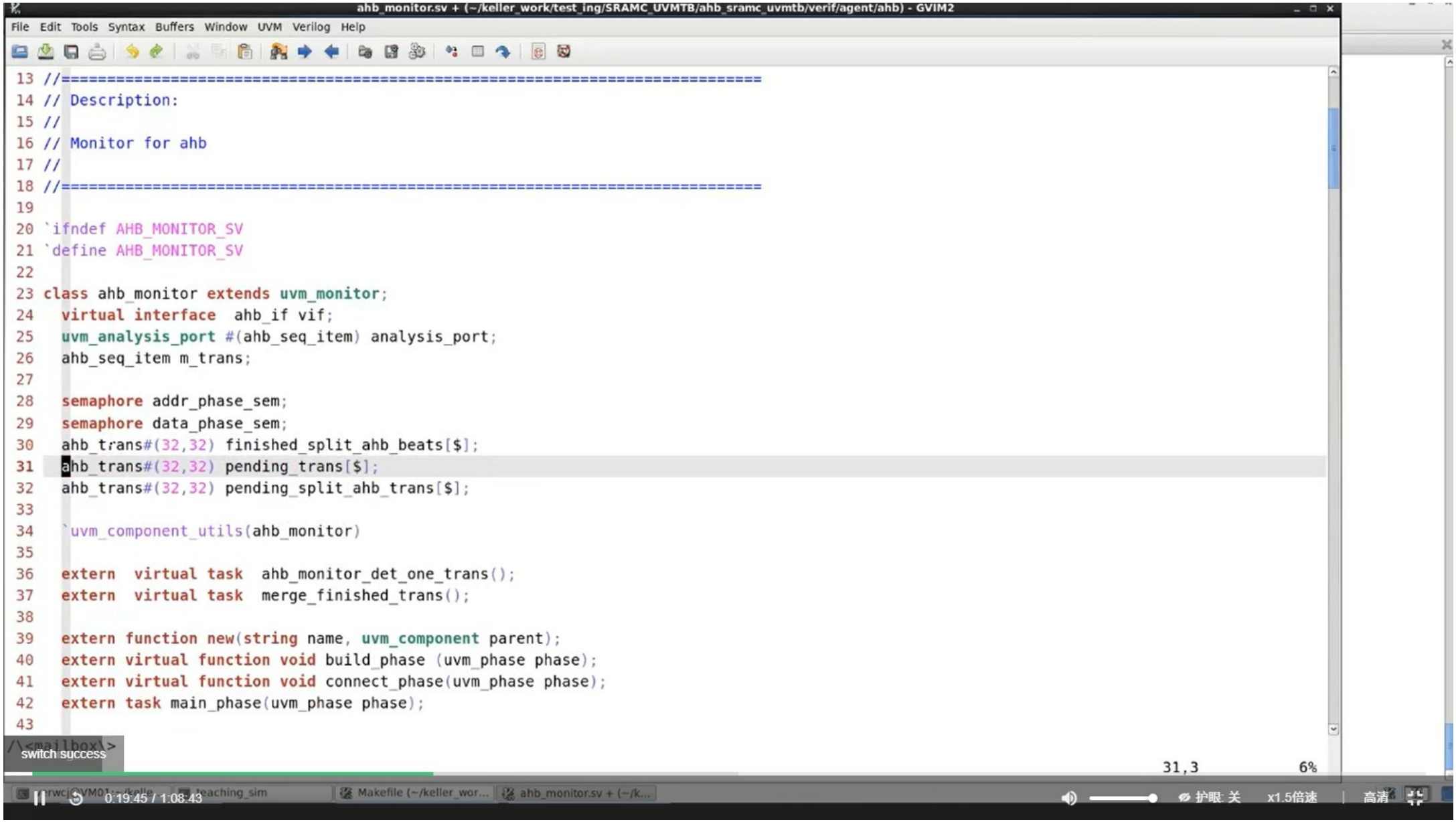

ahb_monitor.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
128 task ahb_monitor::merge_finished_trans();
129 ahb_trans #(32,32) temp_trans;
130 ahb_trans #(32,32) merge_trans;
131 ahb_trans #(32,32) merge_trans_copy;
132 ahb_trans #(32,32) temp_trans_copy;
133 int burst_len = 0;
134 forever begin
135 if(finished_split_ahb_beats.size()>0) begin
136 temp_trans = finished_split_ahb_beats.pop_front();
137 if(temp_trans.trans == IDLE) begin
138 if(merge_trans != null) begin//must not == !null
139 $display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
140 merge_trans.print();
141 //finished_trans.push_back(merge_trans);
142 merge_trans_copy = new merge_trans;
143 analysis_port.write(merge_trans_copy);
144 merge_trans = null;
145 end
146 //finished_trans.push_back(temp_trans);
147 $display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
148 temp_trans.print();
149 temp_trans_copy = new temp_trans;
150 mon2scb.put(temp_trans_copy);
151 end else if(temp_trans.trans == BUSY) begin
152 end else if(temp_trans.trans == NONSEQ) begin
153 //$display("[AHB_MONITOR] @%0t: AHB master driver come into NONSEQ merge.", $time);
154 //temp_trans.print();
155 if(merge_trans != null) begin
156 $display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
157 merge_trans.print();
158 //finished_trans.push_back(merge_trans);
"agent/ahb/ahb_monitor.sv" 221L, 7998C written
-- VISUAL --
143,33 66%
www.coreshinelearning.com userhd7j05
switch success
0:17:00 / 1:08:43
x1.5倍速
高清

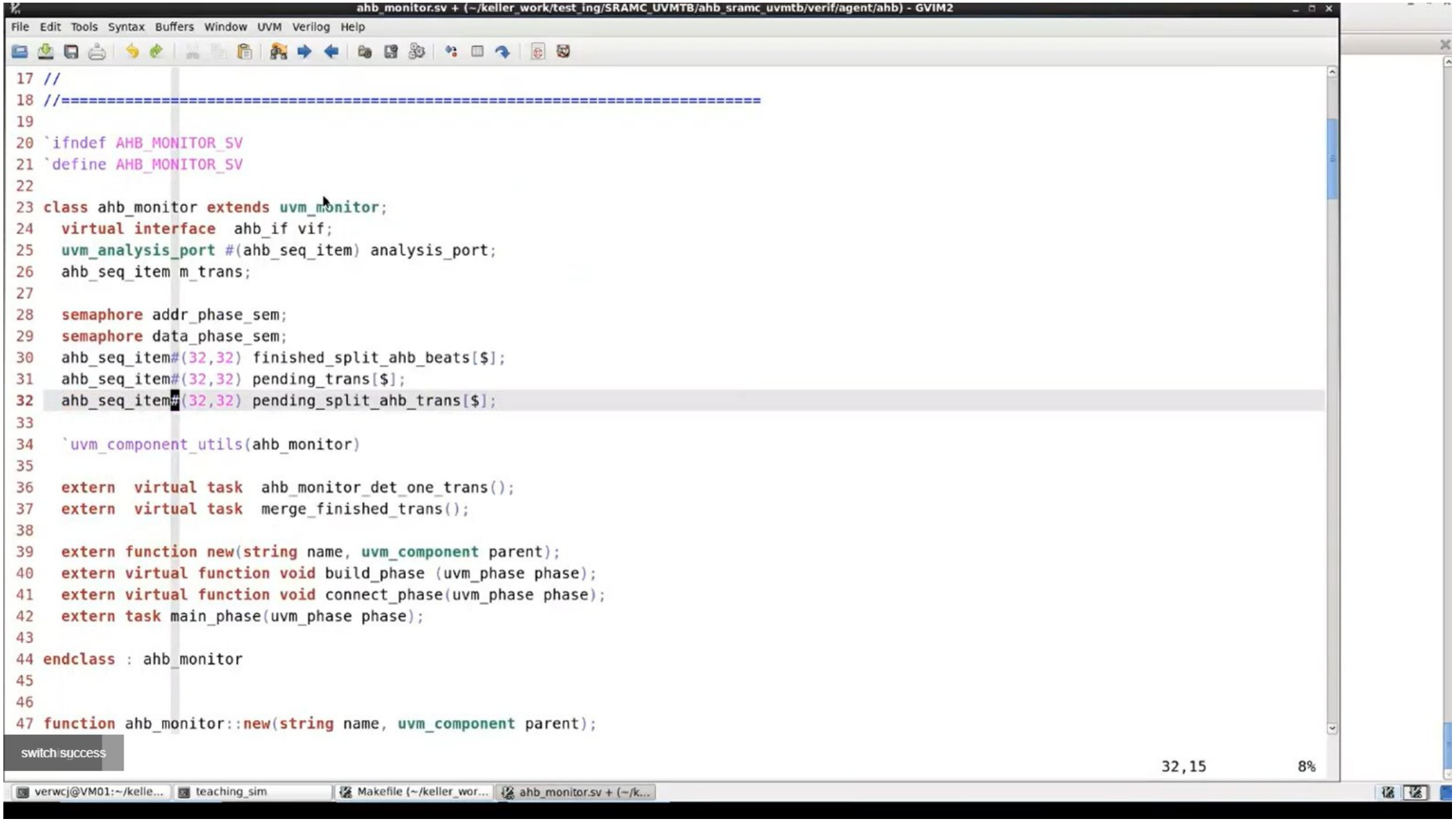

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
merge_trans.print();
//finished_trans.push_back(merge_trans);
merge_trans_copy = new merge_trans;
analysis_port.write(merge_trans_copy);
merge_trans = null;
end
//finished_trans.push_back(temp_trans);
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
temp_trans.print();
temp_trans_copy = new temp_trans;
analysis_port.write(temp_trans_copy);
end else if(temp_trans.trans == BUSY) begin
end else if(temp_trans.trans == NONSEQ) begin
//$display("[AHB_MONITOR] @%0t: AHB master driver come into NONSEQ merge.", $time);
//temp_trans.print();
if(merge_trans != null) begin
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
merge_trans.print();
//finished_trans.push_back(merge_trans);
merge_trans_copy = new merge_trans;
mon2scb.put(merge_trans_copy);
merge_trans = null;
end
merge_trans = new();
merge_trans.haddr = temp_trans.haddr;
merge_trans.beats_addr.push_back(temp_trans.haddr);
merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
merge_trans.trans = temp_trans.trans;
merge_trans.op = temp_trans.op ;
merge_trans.burst = temp_trans.burst;
www.coreshinelearning.com userhd7y05
switch success
160,15 72%
0:17:10 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

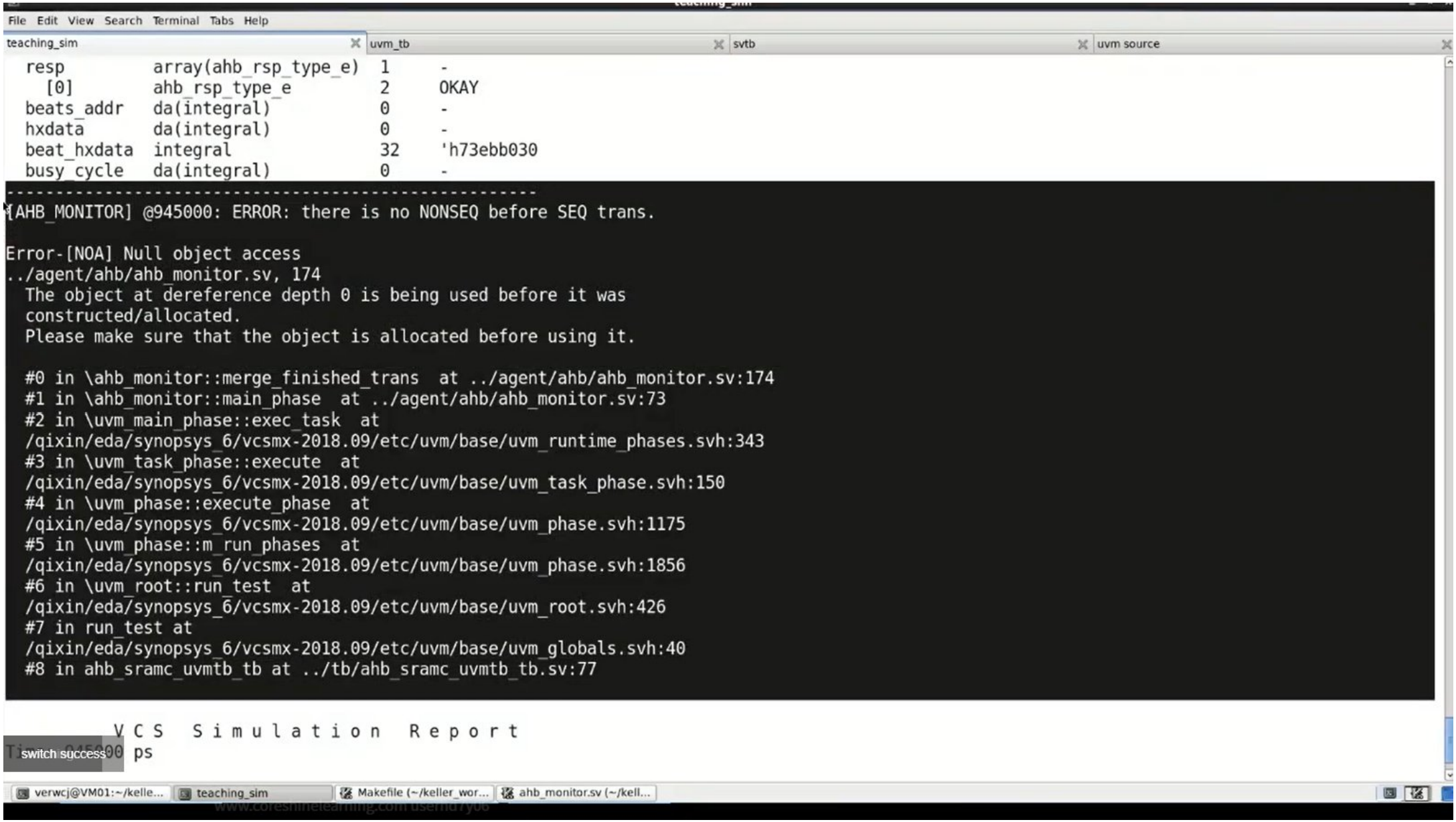

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
www.coreshinelearning.com userhd7y06
File Edit Tools Syntax Buffers Window UVM Verilog Help
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
merge_trans.print();
//finished_trans.push_back(merge_trans);
merge_trans_copy = new merge_trans;
analysis_port.write(merge_trans_copy);
merge_trans = null;
end
//finished_trans.push_back(temp_trans);
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
temp_trans.print();
temp_trans_copy = new temp_trans;
analysis_port.write(temp_trans_copy);
end else if(temp_trans.trans == BUSY) begin
end else if(temp_trans.trans == NONSEQ) begin
//$display("[AHB_MONITOR] @%0t: AHB master driver come into NONSEQ merge.", $time);
//temp_trans.print();
if(merge_trans != null) begin
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
merge_trans.print();
//finished_trans.push_back(merge_trans);
merge_trans_copy = new merge_trans;
analysis_port.write(merge_trans_copy);
merge_trans = null;
end
merge_trans = new();
merge_trans.haddr = temp_trans.haddr;
merge_trans.beats_addr.push_back(temp_trans.haddr);
merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
merge_trans.trans = temp_trans.trans;
merge_trans.op = temp_trans.op ;
merge_trans.burst = temp_trans.burst;
switch success
160,34 72%
0:17:15 / 1:08:43
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
护眼: 关
x1.5倍速
高清

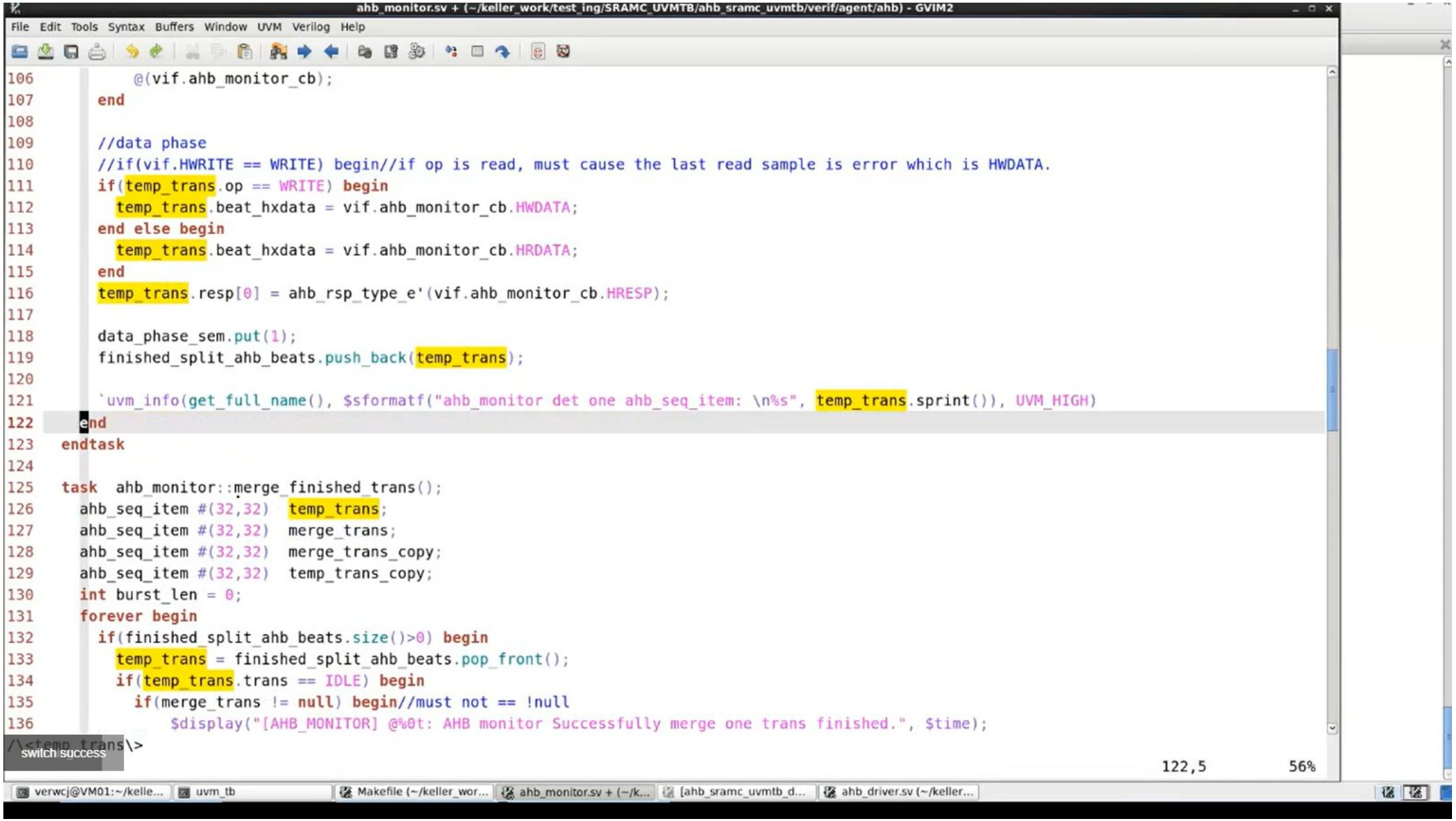

ahb_monitor.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
15 //
16 // Monitor for ahb
17 //
18 //==
19
20 `ifndef AHB_MONITOR_SV
21 `define AHB_MONITOR_SV
22
23 class ahb_monitor extends uvm_monitor;
24 virtual interface ahb_if vif;
25 uvm_analysis_port #(ahb_seq_item) analysis_port;
26 ahb_seq_item m_trans;
27
28 semaphore addr_phase_sem;
29 semaphore data_phase_sem;
30 ahb_trans#(32,32) finished_split_ahb_beats[$];
31 mailbox #(ahb_trans#(32,32)) gen2drv;
32 ahb_trans#(32,32) pending_trans[$];
33 ahb_trans#(32,32) pending_split_ahb_trans[$];
34
35 `uvm_component_utils(ahb_monitor)
36
37 extern virtual task ahb_monitor_det_one_trans();
38 extern virtual task merge_finished_trans();
39
40 extern function new(string name, uvm_component parent);
41 extern virtual function void build_phase (uvm_phase phase);
42 extern virtual function void connect_phase(uvm_phase phase);
43 extern task main_phase(uvm_phase phase);
44
45 endclass : ahb_monitor
"agent/ahb/ahb_monitor.sv" 220L, 7980C written
switch success
30,3 7%
verwcj@VM01:~/kelle...
svtb
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...

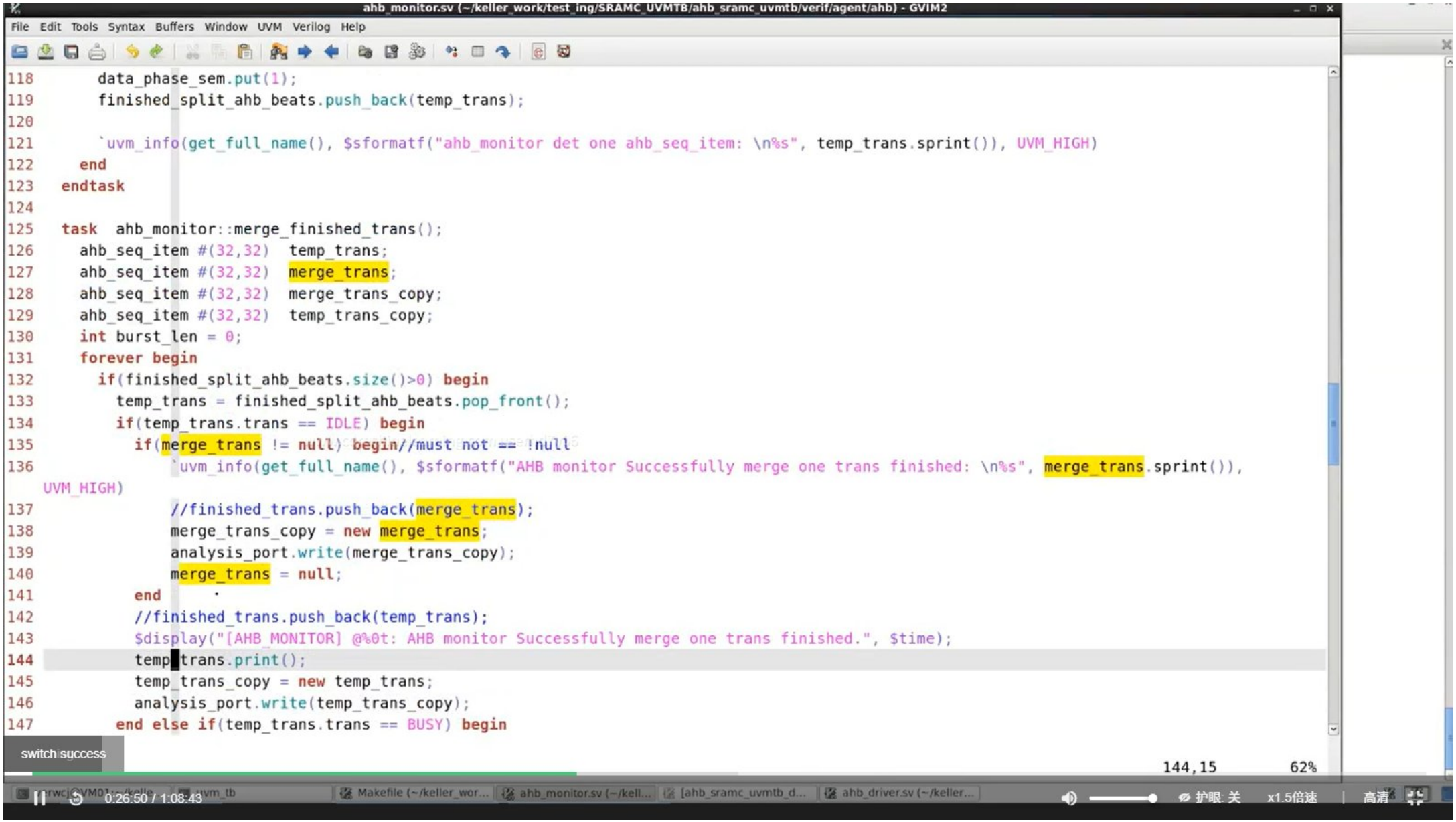

File Edit View Search Terminal Tabs Help
teaching_sim
uvm_tb
svtb
uvm source
Parsing included file '/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/verdi/uvm_verdi_pli.svh'.
Back to file '/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/verdi/uvm_custom_install_verdi_recorder.sv'.
Parsing design file '/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/verdi/uvm_custom_install_verdi_recorder.sv'
Parsing design file '../../rtl/sram_50m_8k_m16/bt_sram_8kx8.v'
Parsing design file '../../rtl/sram_50m_8k_m16/sram_bist.v'
Parsing design file '../../rtl/ahb_sramc/ahb_slave_if.v'
Parsing design file '../../rtl/ahb_sramc/sram_core.v'
Parsing design file '../../rtl/ahb_sramc/sramc_top.v'
Parsing design file '../agent/ahb/ahb_pkg.sv'
Parsing included file '/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/uvm_macros.svh'.
Back to file '../agent/ahb/ahb_pkg.sv'.
Parsing included file '../agent/ahb/ahb_seq_item.sv'.
Back to file '../agent/ahb/ahb_pkg.sv'.
Parsing included file '../agent/ahb/ahb_agent_config.sv'.
Back to file '../agent/ahb/ahb_pkg.sv'.
Parsing included file '../agent/ahb/ahb_monitor.sv'.
Error-[SE] Syntax error
Following verilog source has syntax error :
"../agent/ahb/ahb_monitor.sv", 30: token is '#'
ahb_trans#(32,32) finished_split_ahb_beats[$];
^
1 error
CPU time: 1.791 seconds to compile
make: *** [comp] Error 255
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
switch success
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...

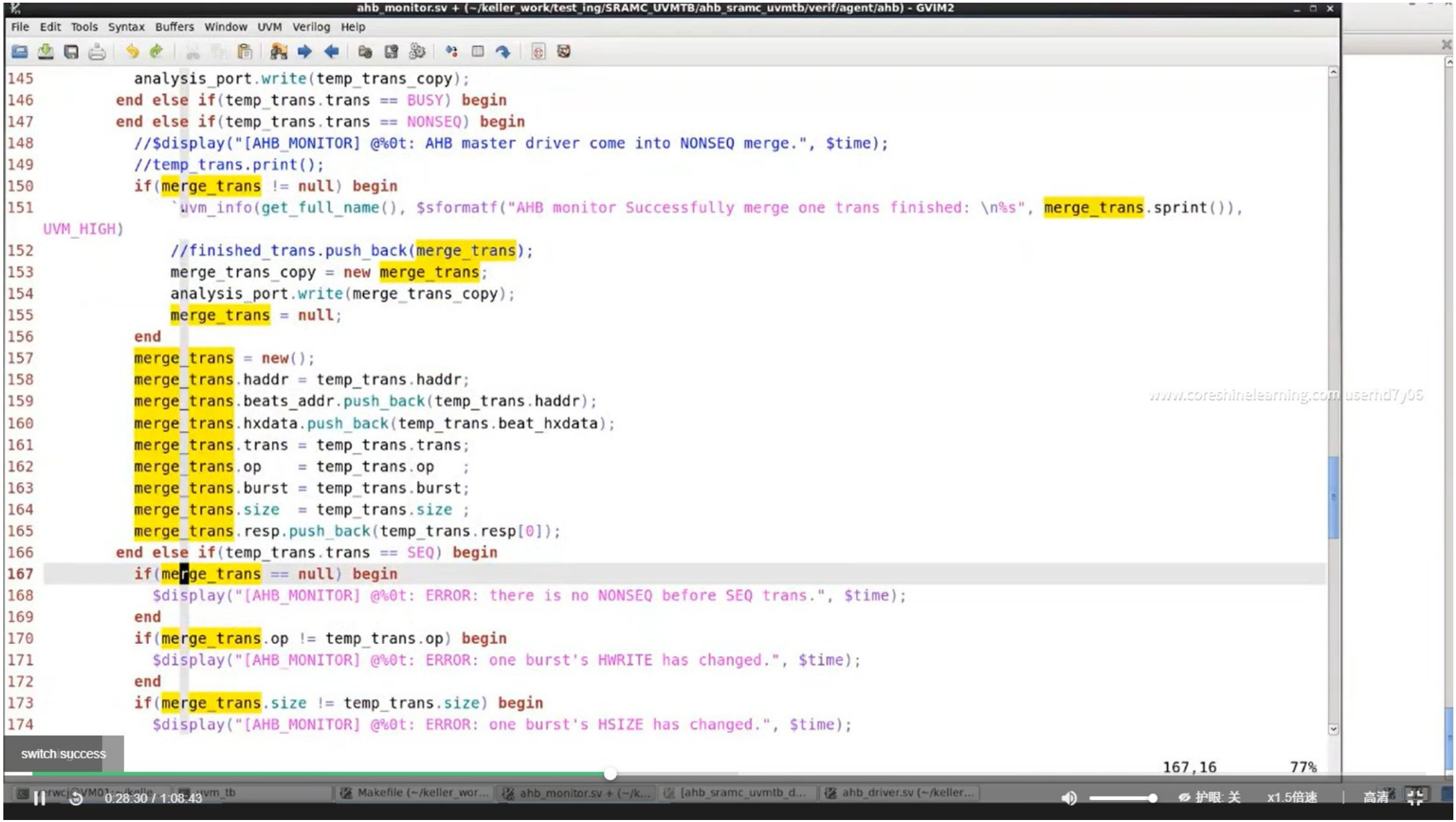

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
13 //==========
14 // Description:
15 //
16 // Monitor for ahb
17 //
18 //==========
19
20 `ifndef AHB_MONITOR_SV
21 `define AHB_MONITOR_SV
22
23 class ahb_monitor extends uvm_monitor;
24 virtual interface ahb_if vif;
25 uvm_analysis_port #(ahb_seq_item) analysis_port;
26 ahb_seq_item m_trans;
27
28 semaphore addr_phase_sem;
29 semaphore data_phase_sem;
30 ahb_trans#(32,32) finished_split_ahb_beats[$];
31 ahb_trans#(32,32) pending_trans[$];
32 ahb_trans#(32,32) pending_split_ahb_trans[$];
33
34 `uvm_component_utils(ahb_monitor)
35
36 extern virtual task ahb_monitor_det_one_trans();
37 extern virtual task merge_finished_trans();
38
39 extern function new(string name, uvm_component parent);
40 extern virtual function void build_phase (uvm_phase phase);
41 extern virtual function void connect_phase(uvm_phase phase);
42 extern task main_phase(uvm_phase phase);
43
/\<mailbox\>
switch success
31,3 6%
0:19:45 / 1:08:43
护眼: 关
x1.5倍速
高清

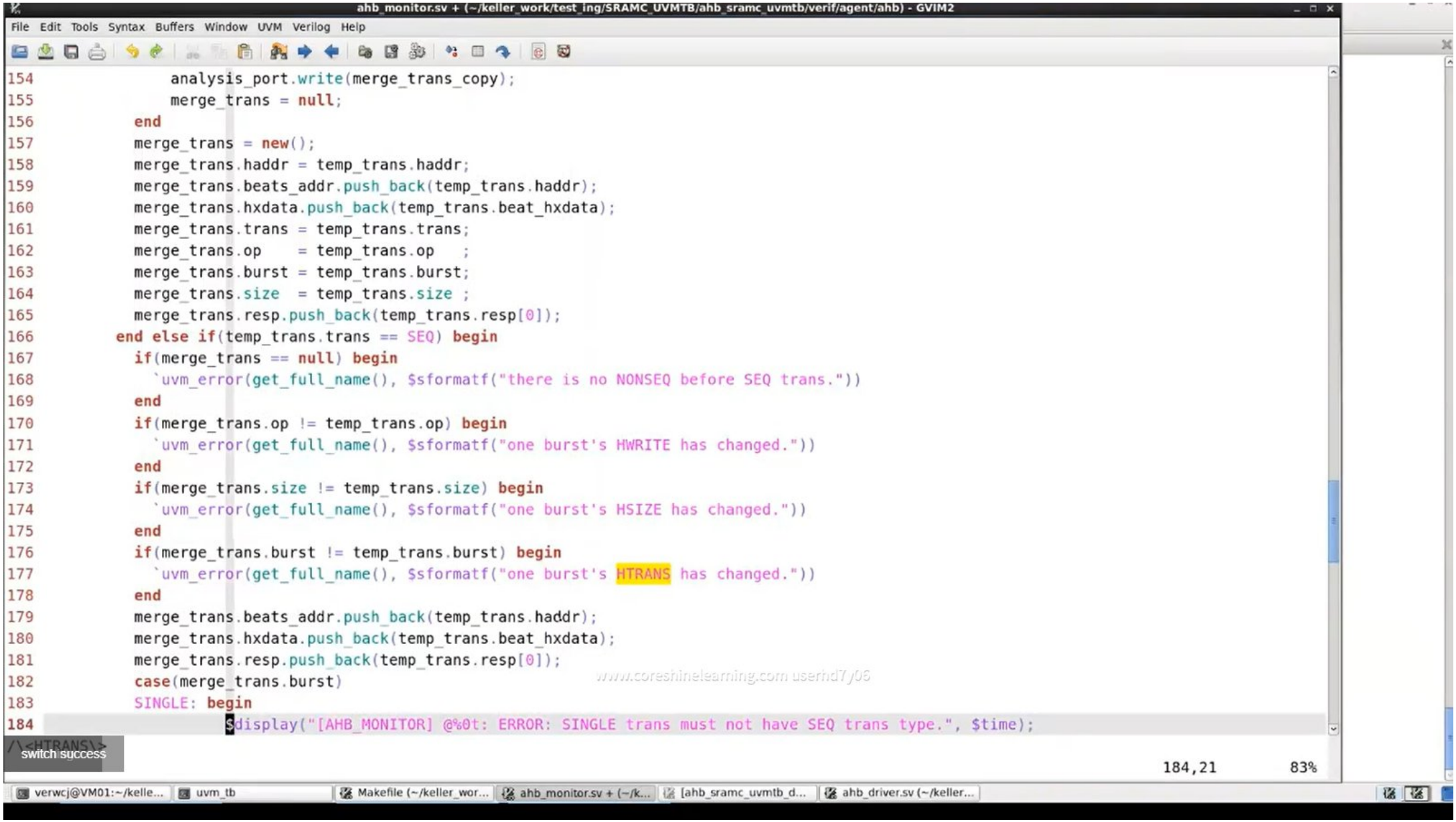

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
17 //
18 //==========
19
20 `ifndef AHB_MONITOR_SV
21 `define AHB_MONITOR_SV
22
23 class ahb_monitor extends uvm_monitor;
24 virtual interface ahb_if vif;
25 uvm_analysis_port #(ahb_seq_item) analysis_port;
26 ahb_seq_item m_trans;
27
28 semaphore addr_phase_sem;
29 semaphore data_phase_sem;
30 ahb_seq_item#(32,32) finished_split_ahb_beats[$];
31 ahb_seq_item#(32,32) pending_trans[$];
32 ahb_seq_item#(32,32) pending_split_ahb_trans[$];
33
34 `uvm_component_utils(ahb_monitor)
35
36 extern virtual task ahb_monitor_det_one_trans();
37 extern virtual task merge_finished_trans();
38
39 extern function new(string name, uvm_component parent);
40 extern virtual function void build_phase (uvm_phase phase);
41 extern virtual function void connect_phase(uvm_phase phase);
42 extern task main_phase(uvm_phase phase);
43
44 endclass : ahb_monitor
45
46
47 function ahb_monitor::new(string name, uvm_component parent);
switch success
32,15 8%
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...

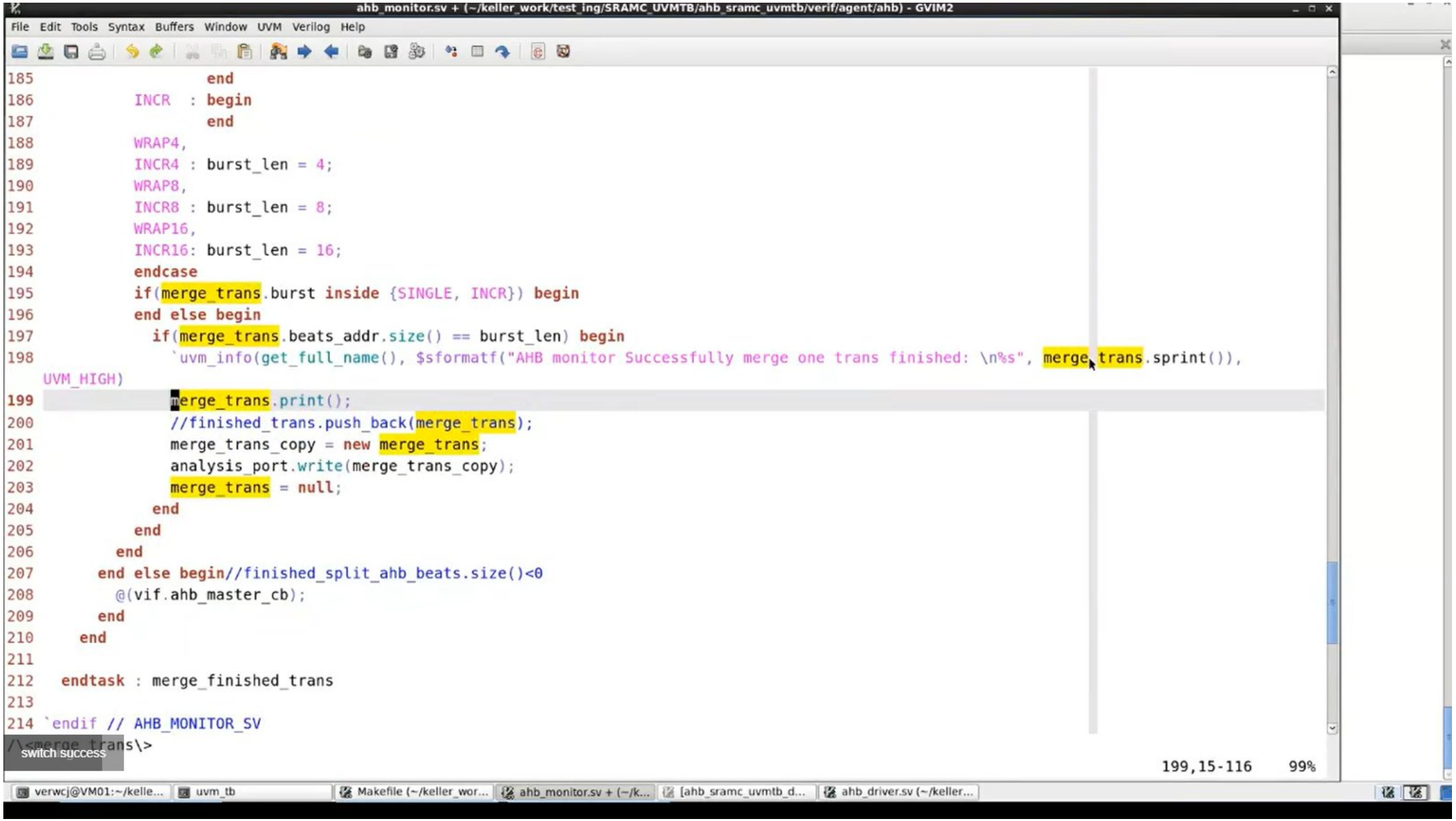

File Edit View Search Terminal Tabs Help
teaching_sim
uvm_tb
svtb
uvm source
resp array(ahb_rsp_type_e) 1 -
[0] ahb_rsp_type_e 2 OKAY
beats_addr da(integral) 0 -
hxdata da(integral) 0 -
beat_hxdata integral 32 'h73ebb030
busy_cycle da(integral) 0 -
[AHB_MONITOR] @945000: ERROR: there is no NONSEQ before SEQ trans.
Error-[NOA] Null object access
../agent/ahb/ahb_monitor.sv, 174
The object at dereference depth 0 is being used before it was
constructed/allocated.
Please make sure that the object is allocated before using it.
#0 in \ahb_monitor::merge_finished_trans at ../agent/ahb/ahb_monitor.sv:174
#1 in \ahb_monitor::main_phase at ../agent/ahb/ahb_monitor.sv:73
#2 in \uvm_main_phase::exec_task at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_runtime_phases.svh:343
#3 in \uvm_task_phase::execute at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_task_phase.svh:150
#4 in \uvm_phase::execute_phase at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_phase.svh:1175
#5 in \uvm_phase::m_run_phases at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_phase.svh:1856
#6 in \uvm_root::run_test at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_root.svh:426
#7 in run_test at
/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_globals.svh:40
#8 in ahb_sramc_uvmtb_tb at ../tb/ahb_sramc_uvmtb_tb.sv:77
V C S S i m u l a t i o n R e p o r t
switch success
ps
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...

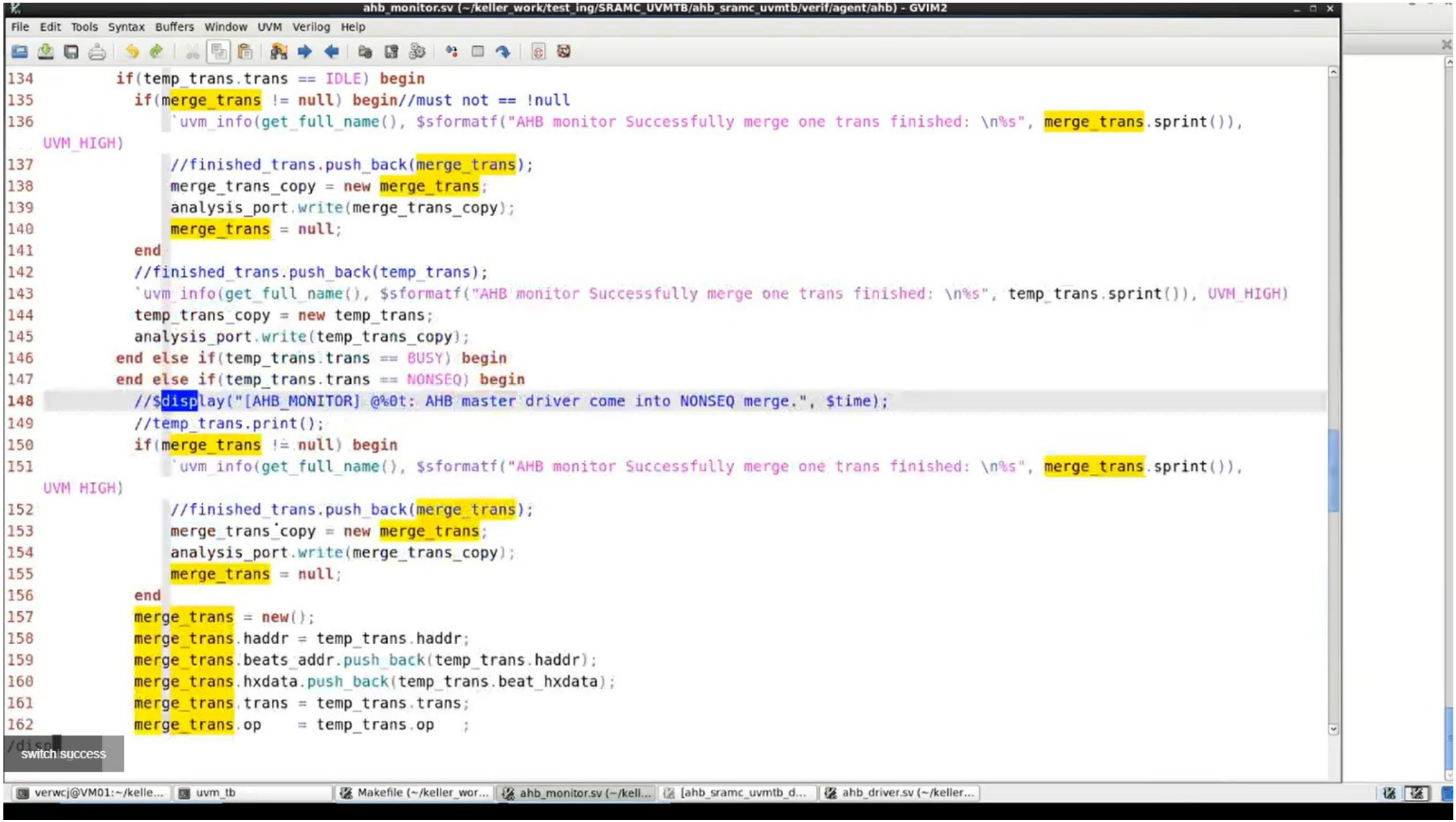

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
@(vif.ahb_monitor_cb);
end
//data phase
//if(vif.HWRITE == WRITE) begin//if op is read, must cause the last read sample is error which is HWDATA.
if(temp_trans.op == WRITE) begin
temp_trans.beat_hxdata = vif.ahb_monitor_cb.HWDATA;
end else begin
temp_trans.beat_hxdata = vif.ahb_monitor_cb.HRDATA;
end
temp_trans.resp[0] = ahb_rsp_type_e'(vif.ahb_monitor_cb.HRESP);
data_phase_sem.put(1);
finished_split_ahb_beats.push_back(temp_trans);
`uvm_info(get_full_name(), $sformatf("ahb_monitor det one ahb_seq_item: \n%s", temp_trans.sprint()), UVM_HIGH)
end
endtask
task ahb_monitor::merge_finished_trans();
ahb_seq_item #(32,32) temp_trans;
ahb_seq_item #(32,32) merge_trans;
ahb_seq_item #(32,32) merge_trans_copy;
ahb_seq_item #(32,32) temp_trans_copy;
int burst_len = 0;
forever begin
if(finished_split_ahb_beats.size()>0) begin
temp_trans = finished_split_ahb_beats.pop_front();
if(temp_trans.trans == IDLE) begin
if(merge_trans != null) begin//must not == !null
$display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
/\<temp_trans\>
switch success
122,5 56%
verwcj@VM01:~/kelle...
uvm_tb
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
[ahb_sramc_uvmtb_d...
ahb_driver.sv (~/keller...

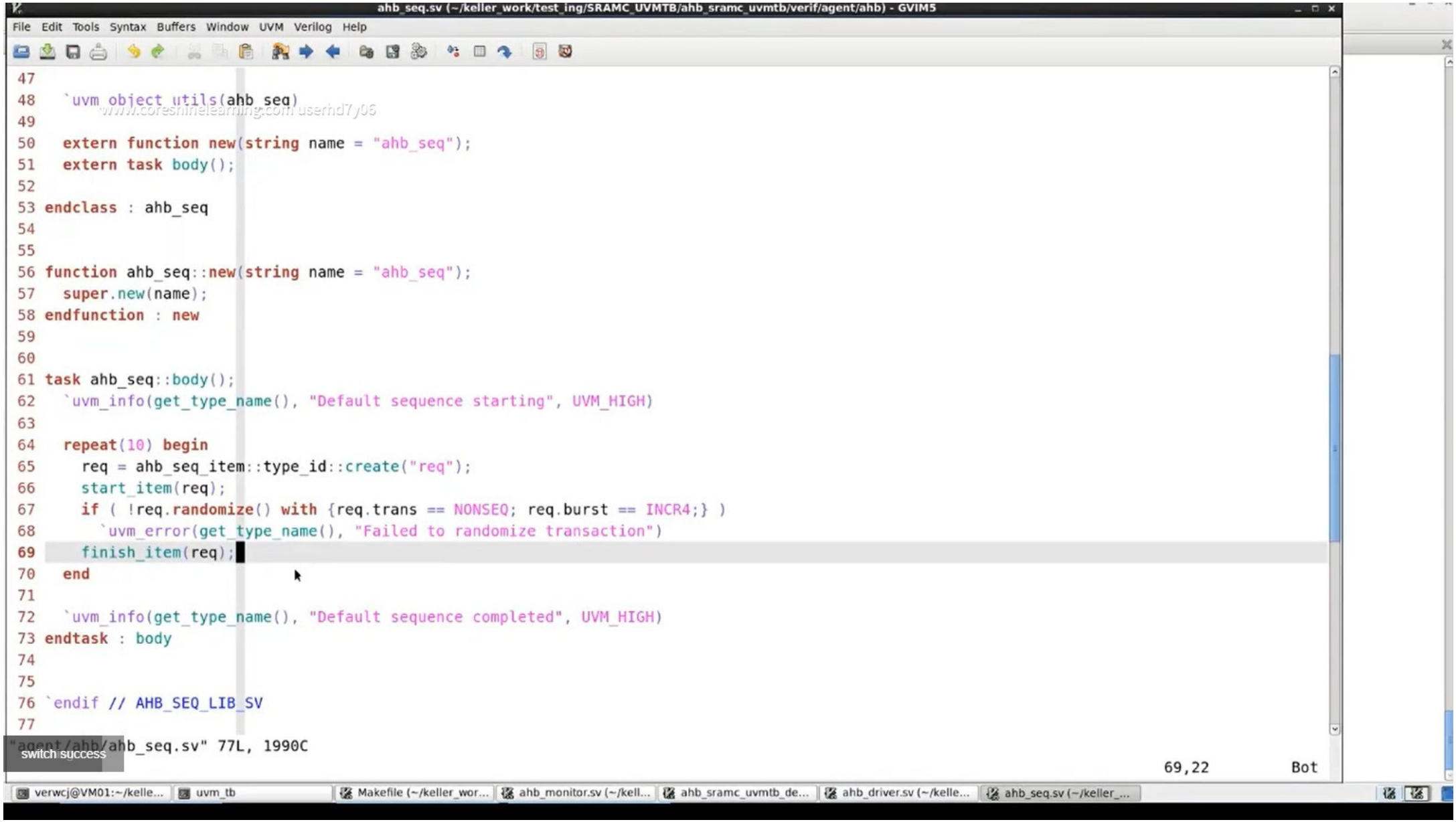

ahb_monitor.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
118 data_phase_sem.put(1);
119 finished_split_ahb_beats.push_back(temp_trans);
120
121 `uvm_info(get_full_name(), $sformatf("ahb_monitor det one ahb_seq_item: \n%s", temp_trans.sprint()), UVM_HIGH)
122 end
123 endtask
124
125 task ahb_monitor::merge_finished_trans();
126 ahb_seq_item #(32,32) temp_trans;
127 ahb_seq_item #(32,32) merge_trans;
128 ahb_seq_item #(32,32) merge_trans_copy;
129 ahb_seq_item #(32,32) temp_trans_copy;
130 int burst_len = 0;
131 forever begin
132 if(finished_split_ahb_beats.size()>0) begin
133 temp_trans = finished_split_ahb_beats.pop_front();
134 if(temp_trans.trans == IDLE) begin
135 if(merge_trans != null) begin//must not == !null
136 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", merge_trans.sprint()), UVM_HIGH)
137 //finished_trans.push_back(merge_trans);
138 merge_trans_copy = new merge_trans;
139 analysis_port.write(merge_trans_copy);
140 merge_trans = null;
141 end
142 //finished_trans.push_back(temp_trans);
143 $display("[AHB_MONITOR] @%0t: AHB monitor Successfully merge one trans finished.", $time);
144 temp_trans.print();
145 temp_trans_copy = new temp_trans;
146 analysis_port.write(temp_trans_copy);
147 end else if(temp_trans.trans == BUSY) begin
switch success
144,15 62%
0:26:50 / 1:08:43
x1.5倍速
高清


ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
145 analysis_port.write(temp_trans_copy);
146 end else if(temp_trans.trans == BUSY) begin
147 end else if(temp_trans.trans == NONSEQ) begin
148 //$display("[AHB_MONITOR] @%0t: AHB master driver come into NONSEQ merge.", $time);
149 //temp_trans.print();
150 if(merge_trans != null) begin
151 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", merge_trans.sprint()), UVM_HIGH)
152 //finished_trans.push_back(merge_trans);
153 merge_trans_copy = new merge_trans;
154 analysis_port.write(merge_trans_copy);
155 merge_trans = null;
156 end
157 merge_trans = new();
158 merge_trans.haddr = temp_trans.haddr;
159 merge_trans.beats_addr.push_back(temp_trans.haddr);
160 merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
161 merge_trans.trans = temp_trans.trans;
162 merge_trans.op = temp_trans.op ;
163 merge_trans.burst = temp_trans.burst;
164 merge_trans.size = temp_trans.size ;
165 merge_trans.resp.push_back(temp_trans.resp[0]);
166 end else if(temp_trans.trans == SEQ) begin
167 if(merge_trans == null) begin
168 $display("[AHB_MONITOR] @%0t: ERROR: there is no NONSEQ before SEQ trans.", $time);
169 end
170 if(merge_trans.op != temp_trans.op) begin
171 $display("[AHB_MONITOR] @%0t: ERROR: one burst's HWRITE has changed.", $time);
172 end
173 if(merge_trans.size != temp_trans.size) begin
174 $display("[AHB_MONITOR] @%0t: ERROR: one burst's HSIZE has changed.", $time);
switch success
167,16 77%
www.coreshinelearning.com userid7y06
0:28:30 / 1:08:43
x1.5倍速
高清

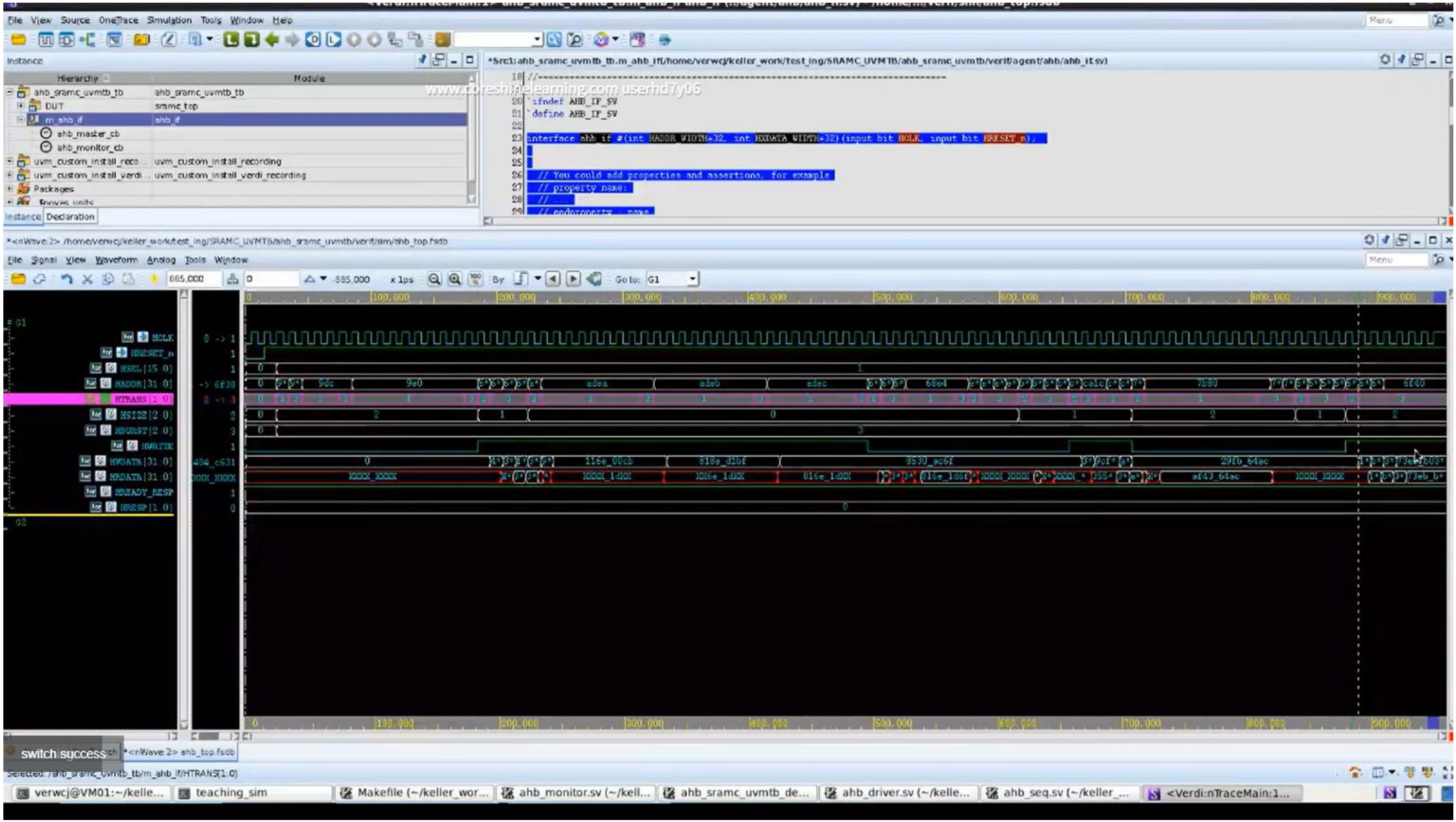

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
analysis_port.write(merge_trans_copy);
merge_trans = null;
end
merge_trans = new();
merge_trans.haddr = temp_trans.haddr;
merge_trans.beats_addr.push_back(temp_trans.haddr);
merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
merge_trans.trans = temp_trans.trans;
merge_trans.op = temp_trans.op ;
merge_trans.burst = temp_trans.burst;
merge_trans.size = temp_trans.size ;
merge_trans.resp.push_back(temp_trans.resp[0]);
end else if(temp_trans.trans == SEQ) begin
if(merge_trans == null) begin
`uvm_error(get_full_name(), $sformatf("there is no NONSEQ before SEQ trans."))
end
if(merge_trans.op != temp_trans.op) begin
`uvm_error(get_full_name(), $sformatf("one burst's HWRITE has changed."))
end
if(merge_trans.size != temp_trans.size) begin
`uvm_error(get_full_name(), $sformatf("one burst's HSIZE has changed."))
end
if(merge_trans.burst != temp_trans.burst) begin
`uvm_error(get_full_name(), $sformatf("one burst's HTRANS has changed."))
end
merge_trans.beats_addr.push_back(temp_trans.haddr);
merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
merge_trans.resp.push_back(temp_trans.resp[0]);
case(merge_trans.burst)
SINGLE: begin
$display("[AHB_MONITOR] @%0t: ERROR: SINGLE trans must not have SEQ trans type.", $time);
/\<HTRANS\>
switch success
184,21 83%
verwcj@VM01:~/kelle... uvm_tb Makefile (~/keller_wor... ahb_monitor.sv + (~/k... [ahb_sramc_uvmtb_d... ahb_driver.sv (~/keller...

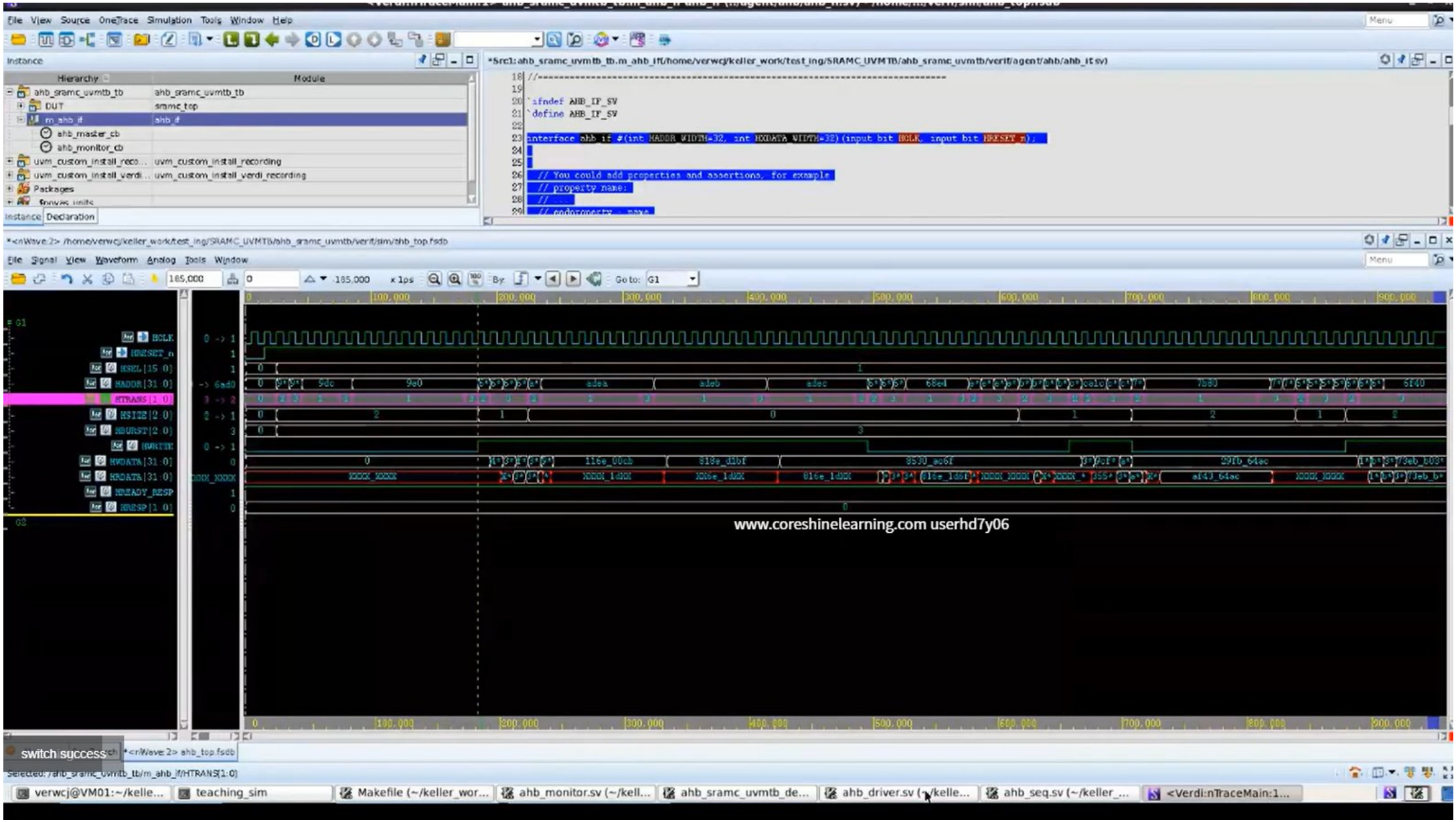

ahb_monitor.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
185 end
186 INCR : begin
187 end
188 WRAP4,
189 INCR4 : burst_len = 4;
190 WRAP8,
191 INCR8 : burst_len = 8;
192 WRAP16,
193 INCR16: burst_len = 16;
194 endcase
195 if(merge_trans.burst inside {SINGLE, INCR}) begin
196 end else begin
197 if(merge_trans.beats_addr.size() == burst_len) begin
198 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", merge_trans.sprint()), UVM_HIGH)
199 merge_trans.print();
200 //finished_trans.push_back(merge_trans);
201 merge_trans_copy = new merge_trans;
202 analysis_port.write(merge_trans_copy);
203 merge_trans = null;
204 end
205 end
206 end
207 end else begin//finished_split_ahb_beats.size()<0
208 @(vif.ahb_master_cb);
209 end
210 end
211
212 endtask : merge_finished_trans
213
214 `endif // AHB_MONITOR_SV
/\<merge_trans\>
switch success
199,15-116 99%
verwcj@VM01:~/kelle...
uvm_tb
Makefile (~/keller_wor...
ahb_monitor.sv + (~/k...
[ahb_sramc_uvmtb_d...
ahb_driver.sv (~/keller...

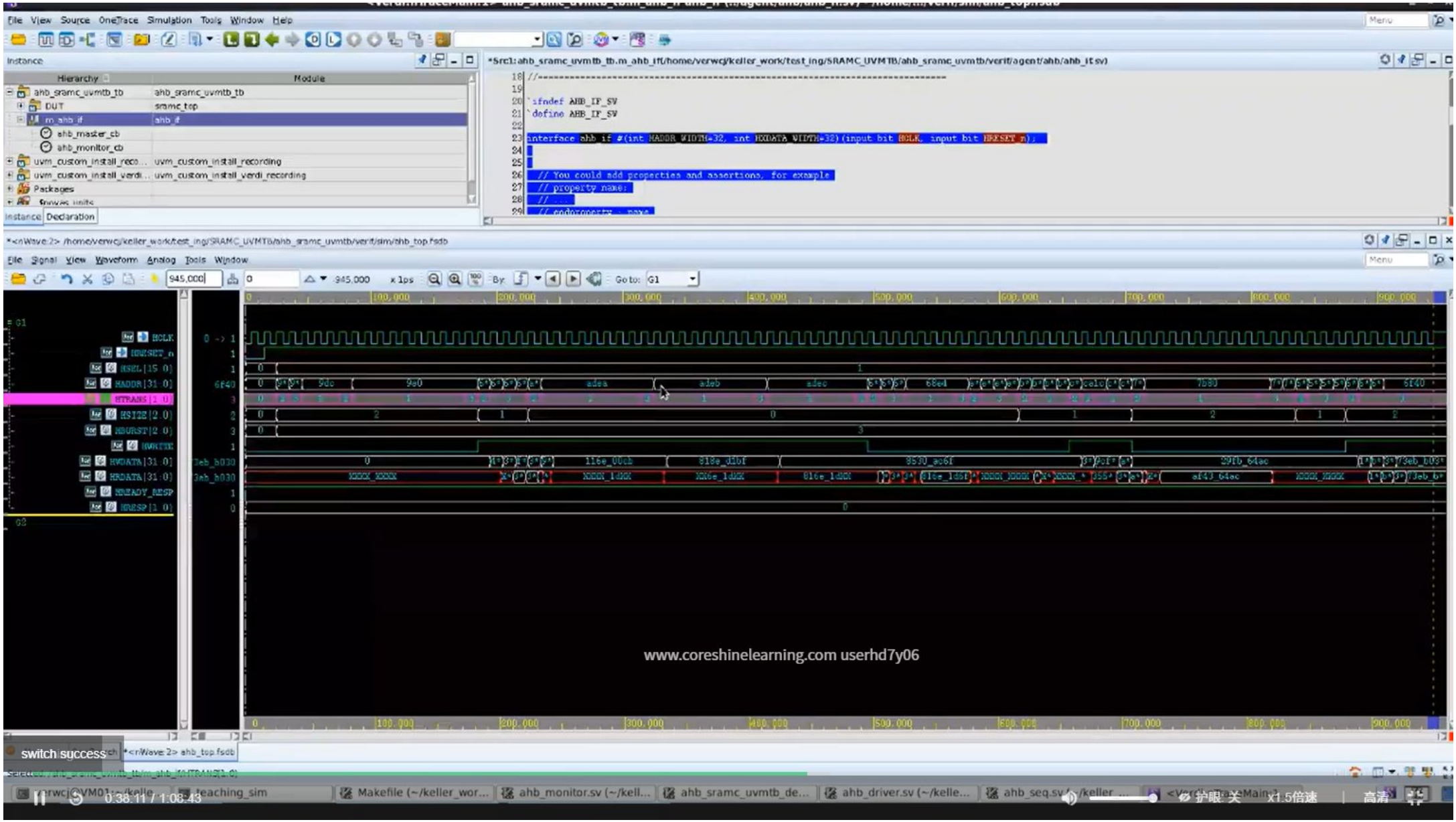

ahb_monitor.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM2
File Edit Tools Syntax Buffers Window UVM Verilog Help
134 if(temp_trans.trans == IDLE) begin
135 if(merge_trans != null) begin//must not == !null
136 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", merge_trans.sprint()), UVM_HIGH)
137 //finished_trans.push_back(merge_trans);
138 merge_trans_copy = new merge_trans;
139 analysis_port.write(merge_trans_copy);
140 merge_trans = null;
141 end
142 //finished_trans.push_back(temp_trans);
143 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", temp_trans.sprint()), UVM_HIGH)
144 temp_trans_copy = new temp_trans;
145 analysis_port.write(temp_trans_copy);
146 end else if(temp_trans.trans == BUSY) begin
147 end else if(temp_trans.trans == NONSEQ) begin
148 //$display("[AHB_MONITOR] @%0t: AHB master driver come into NONSEQ merge.", $time);
149 //temp_trans.print();
150 if(merge_trans != null) begin
151 `uvm_info(get_full_name(), $sformatf("AHB monitor Successfully merge one trans finished: \n%s", merge_trans.sprint()), UVM_HIGH)
152 //finished_trans.push_back(merge_trans);
153 merge_trans_copy = new merge_trans;
154 analysis_port.write(merge_trans_copy);
155 merge_trans = null;
156 end
157 merge_trans = new();
158 merge_trans.haddr = temp_trans.haddr;
159 merge_trans.beats_addr.push_back(temp_trans.haddr);
160 merge_trans.hxdata.push_back(temp_trans.beat_hxdata);
161 merge_trans.trans = temp_trans.trans;
162 merge_trans.op = temp_trans.op ;
/disp
switch success
verwcj@VM01:~/kelle... uvm_tb Makefile (~/keller_wor... ahb_monitor.sv (~/kell... [ahb_sramc_uvmtb_d... ahb_driver.sv (~/keller...

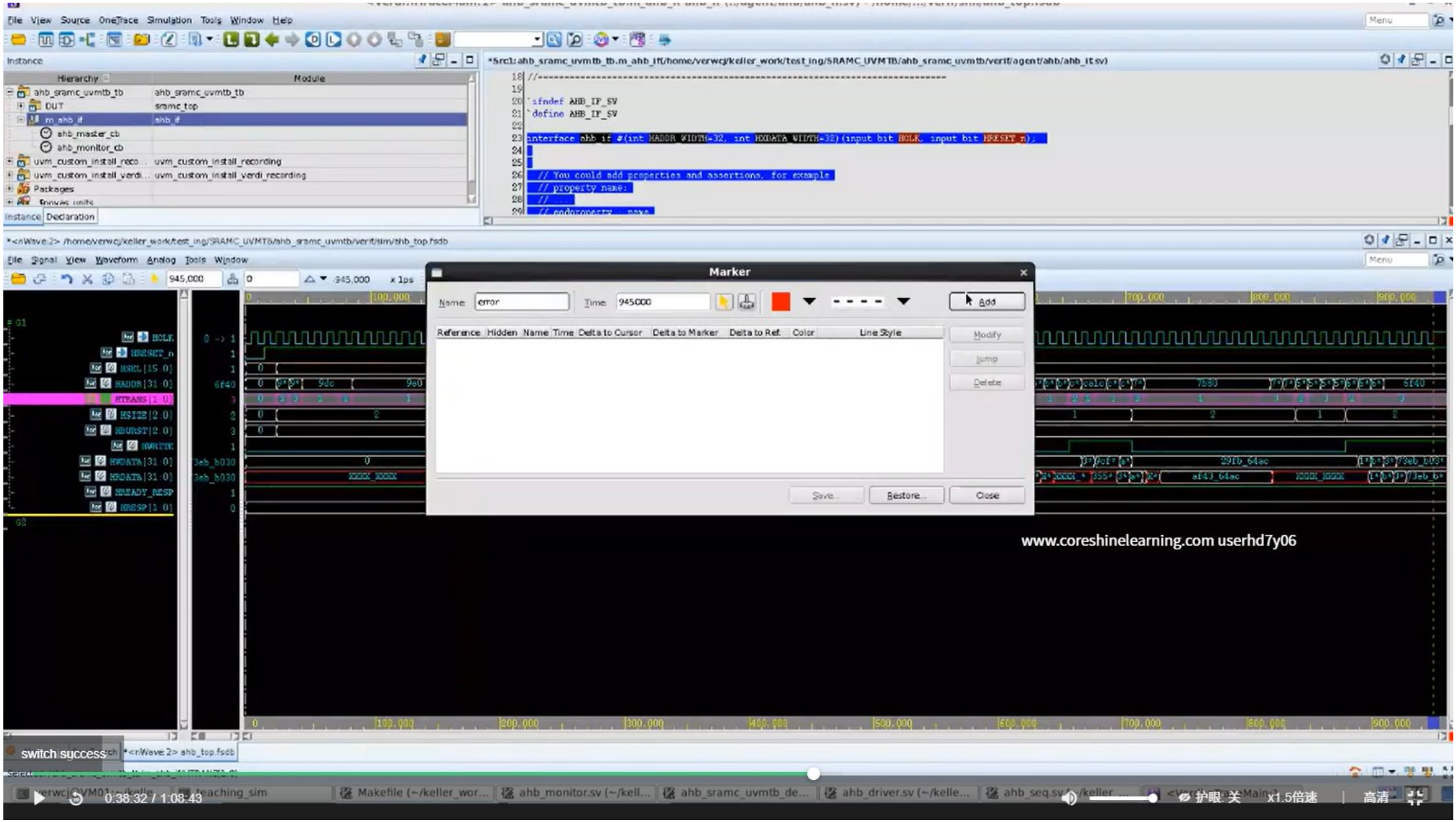


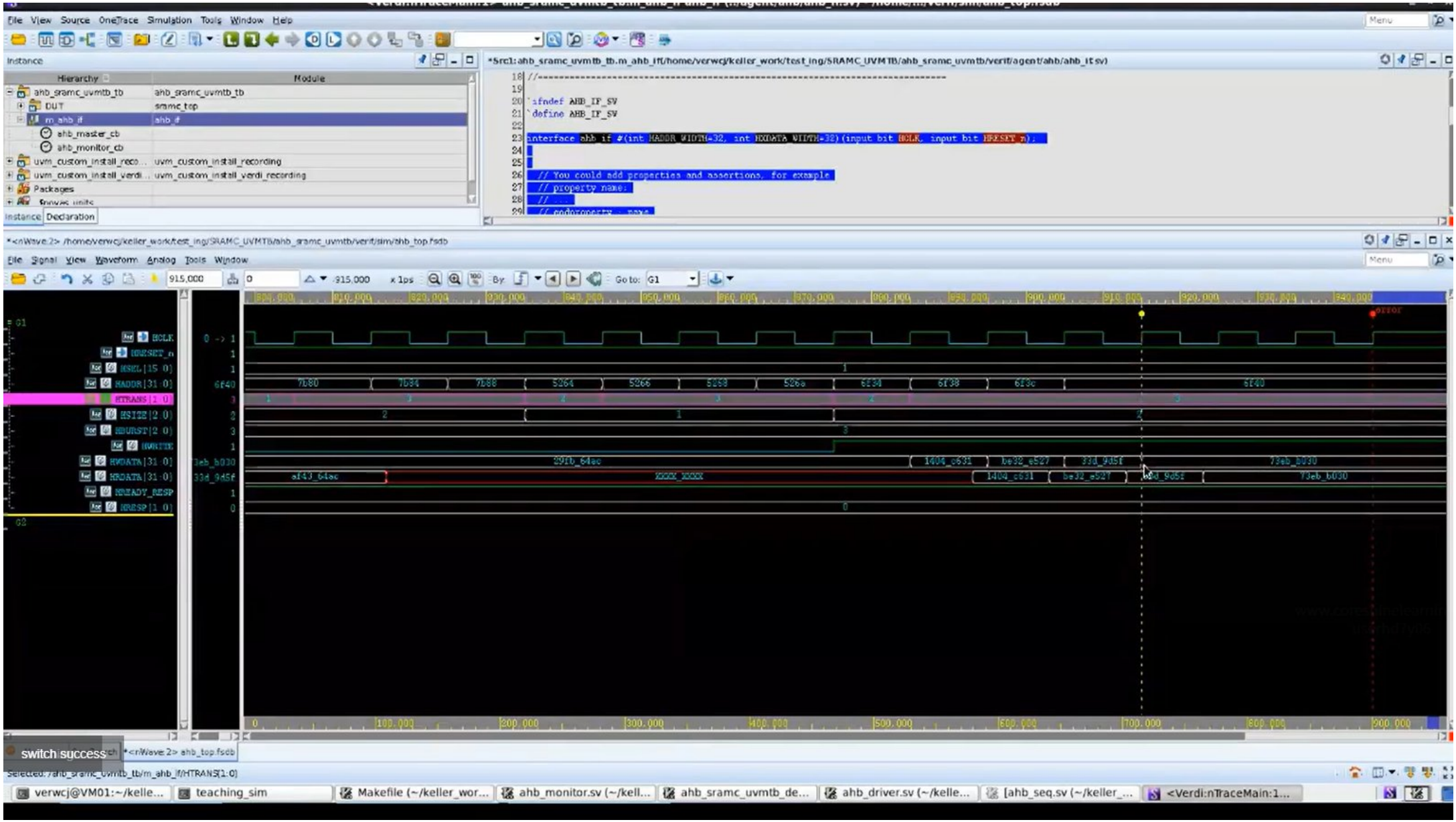
www.coreshinelearning.com/user/id7706
switch success

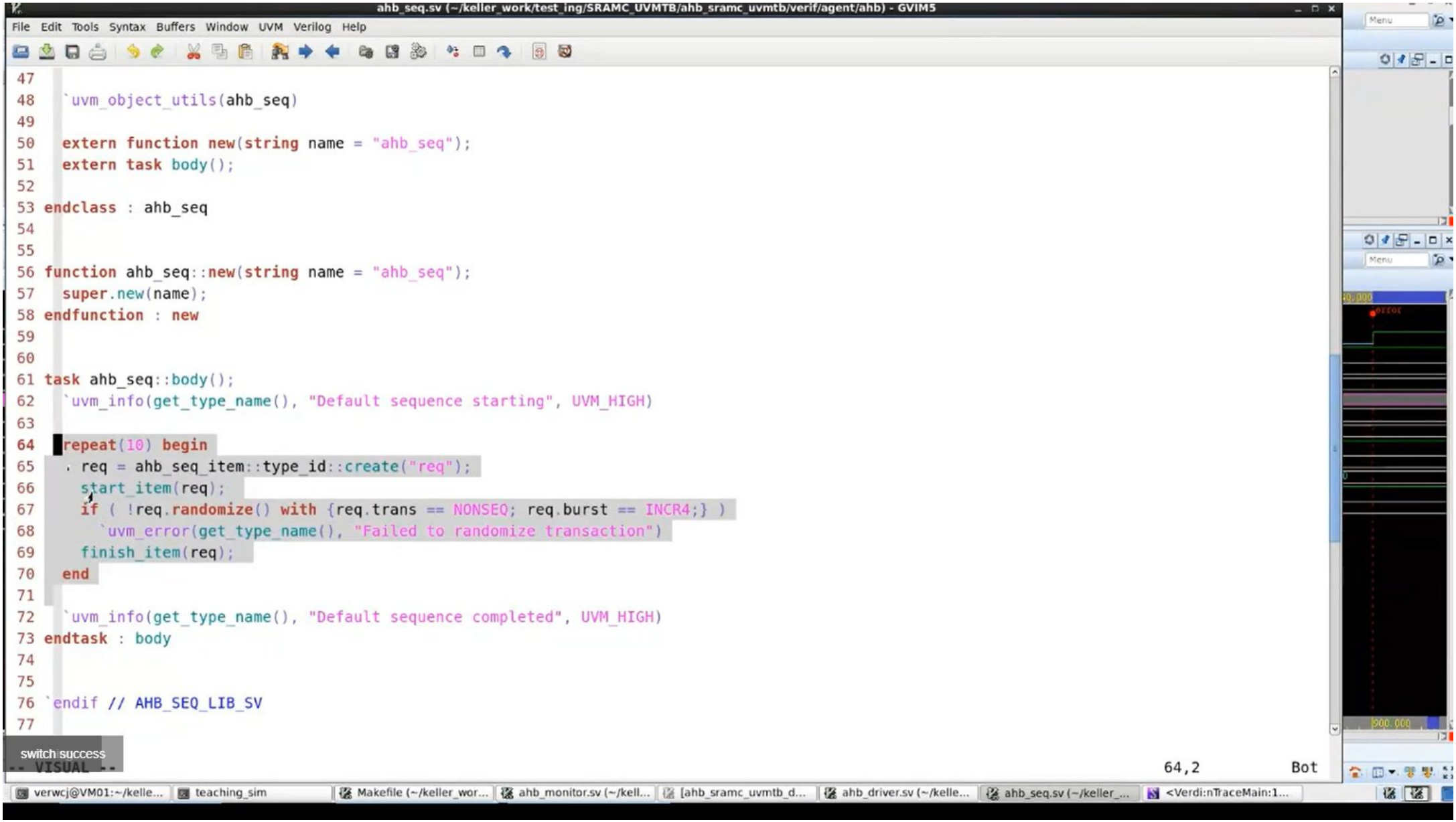

File View Source OneTrace Simulation Tools Window Help
Instance
Hierarchy
Module
ahb_sramc_uvmtb_tb
ahb_sramc_uvmtb_tb
DUT
sramc_top
m_ahb_if
ahb_if
ahb_master_cb
ahb_monitor_cb
uvm_custom_install_reco...
uvm_custom_install_recording
uvm_custom_install_verdi...
uvm_custom_install_verdi_recording
Packages
Instance
Declaration
*Src1:ahb_sramc_uvmtb_tb.m_ahb_if/home/verwcj/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb/ahb_if.sv)
www.coreshinelearning.com userid7y06
`ifndef AHB_IF_SV
`define AHB_IF_SV
interface ahb_if #(int HADDR_WIDTH=32, int HDATA_WIDTH=32)(input bit HCLK, input bit HRESET_n);
// You could add properties and assertions, for example
// property name:
// ...
*<nWave 2> /home/verwcj/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim/ahb_top.fsdb
File Signal View Waveform Analog Tools Window
Go to:
HCLK
HRESET_n
HSEL[15:0]
HADDR[31:0]
HTRANS[1:0]
HSIZE[2:0]
HBURST[2:0]
HWRITE
HWDATA[31:0]
HRDATA[31:0]
HREADY_RESP
HRESP[1:0]
switch success
Selected: /ahb_sramc_uvmtb_tb/m_ahb_if/HTRANS[1:0]
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...
ahb_sramc_uvmtb_de...
ahb_driver.sv (~/kelle...
ahb_seq.sv (~/keller_...
<Verdi:nTraceMain:1...

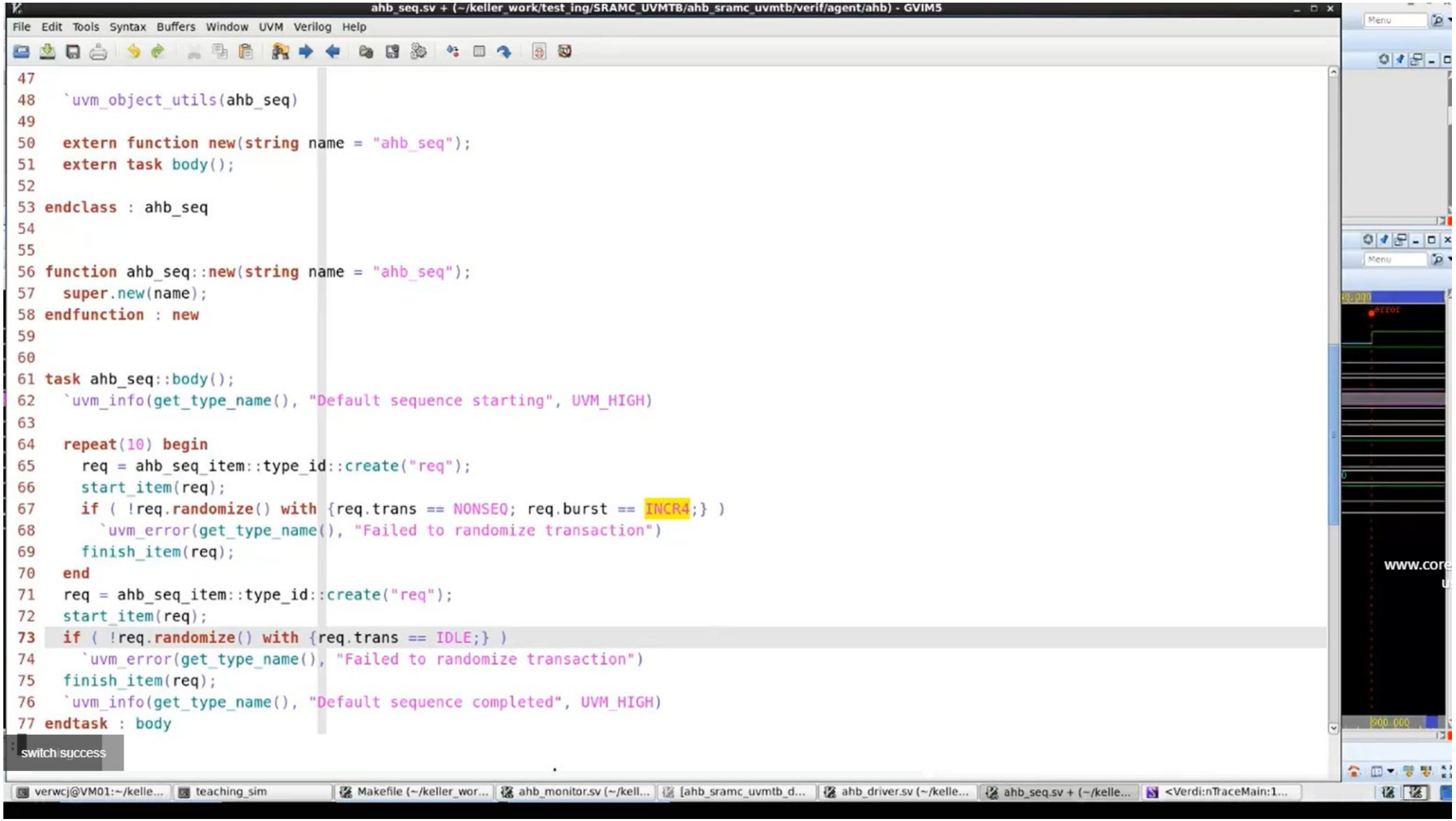

File View Source OneTrace Simulation Tools Window Help
Instance
Hierarchy
Module
ahb_sramc_uvmtb_tb
ahb_sramc_uvmtb_tb
DUT
sramc_top
m_ahb_if
ahb_if
ahb_master_cb
ahb_monitor_cb
uvm_custom_install_reco...
uvm_custom_install_recording
uvm_custom_install_verdi...
uvm_custom_install_verdi_recording
Packages
Instance
Declaration
*Src1:ahb_sramc_uvmtb_tb.m_ahb_if:/home/verwcj/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb/ahb_if.sv)
//
`ifndef AHB_IF_SV
`define AHB_IF_SV
interface ahb_if #(int HADDR_WIDTH=32, int HDDATA_WIDTH=32)(input bit HCLK, input bit HRESET_n);
// You could add properties and assertions, for example
// property name:
File Signal View Waveform Analog Tools Window
Go to: G1
HCLK
HRESET_n
HSEL[15:0]
HADDR[31:0]
HTRANS[1:0]
HSIZE[2:0]
HBURST[2:0]
HWRITE
HWDATA[31:0]
HRDATA[31:0]
HREADY_RESP
HRESP[1:0]
www.coreshinelearning.com userhd7y06
switch success
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...
ahb_sramc_uvmtb_de...
ahb_driver.sv (~/kelle...
ahb_seq.sv (~/keller_...
<Verdi:nTraceMain:1...


File View Source OneTrace Simulation Tools Window Help
Instance
Hierarchy
Module
ahb_sramc_uvmtb_tb
DUT
m_ahb_if
ahb_master_cb
ahb_monitor_cb
Packages
Instance Declaration
*Src1:ahb_sramc_uvmtb_tb.m_ahb_if(/home/verwcj/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb/ahb_if.sv)
`ifndef AHB_IF_SV
`define AHB_IF_SV
interface ahb_if #(int HADDR_WIDTH=32, int HDATA_WIDTH=32)(input bit HCLK, input bit HRESET_n);
// You could add properties and assertions, for example
// property name:
File Signal View Waveform Analog Tools Window
HCLK
HRESET_n
HSEL[15:0]
HADDR[31:0]
HTRANS[1:0]
HSIZE[2:0]
HBURST[2:0]
HWRITE
HWDATA[31:0]
HRDATA[31:0]
HREADY_RESP
HRESP[1:0]
www.coreshinelearning.com userhd7y06
switch success
0:38:11 / 1:08:43
x1.5倍速
高清


Marker
Name: error
Time: 945000
Add
Modify
Jump
Delete
Save...
Restore...
Close
www.coreshinelearning.com userhd7y06
switch success
0:38:32 / 1:08:43
x1.5倍速
高清


File View Source OneTrace Simulation Tools Window Help
Instance
Hierarchy
Module
ahb_sramc_uvmtb_tb
DUT
sramc_top
m_ahb_if
ahb_if
ahb_master_cb
ahb_monitor_cb
Packages
Declaration
*Src1:ahb_sramc_uvmtb_tb.m_ahb_if(/home/verwcj/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb/ahb_if.sv)
//
`ifndef AHB_IF_SV
`define AHB_IF_SV
interface ahb_if #(int HADDR_WIDTH=32, int HDATA_WIDTH=32)(input bit HCLK, input bit HRESET_n);
// You could add properties and assertions, for example
// property name;
File Signal View Waveform Analog Tools Window
Go to:
HCLK
HRESET_n
HSEL[15:0]
HADDR[31:0]
HTRANS[1:0]
HSIZE[2:0]
HBURST[2:0]
HWRITE
HWDATA[31:0]
HRDATA[31:0]
HREADY_RESP
HRESP[1:0]
switch success
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...
ahb_sramc_uvmtb_de...
ahb_driver.sv (~/kelle...
[ahb_seq.sv (~/keller_...
<Verdi:nTraceMain:1...


ahb_seq.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM5
File Edit Tools Syntax Buffers Window UVM Verilog Help
47
48 `uvm_object_utils(ahb_seq)
49
50 extern function new(string name = "ahb_seq");
51 extern task body();
52
53 endclass : ahb_seq
54
55
56 function ahb_seq::new(string name = "ahb_seq");
57 super.new(name);
58 endfunction : new
59
60
61 task ahb_seq::body();
62 `uvm_info(get_type_name(), "Default sequence starting", UVM_HIGH)
63
64 repeat(10) begin
65 req = ahb_seq_item::type_id::create("req");
66 start_item(req);
67 if (!req.randomize() with {req.trans == NONSEQ; req.burst == INCR4;})
68 `uvm_error(get_type_name(), "Failed to randomize transaction")
69 finish_item(req);
70 end
71
72 `uvm_info(get_type_name(), "Default sequence completed", UVM_HIGH)
73 endtask : body
74
75
76 `endif // AHB_SEQ_LIB_SV
77
switch success
-- VISUAL --
64,2 Bot
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...
[ahb_sramc_uvmtb_d...
ahb_driver.sv (~/kelle...
ahb_seq.sv (~/keller_...
<Verdi:nTraceMain:1...


ahb_seq.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM5
File Edit Tools Syntax Buffers Window UVM Verilog Help
47
48 `uvm_object_utils(ahb_seq)
49
50 extern function new(string name = "ahb_seq");
51 extern task body();
52
53 endclass : ahb_seq
54
55
56 function ahb_seq::new(string name = "ahb_seq");
57 super.new(name);
58 endfunction : new
59
60
61 task ahb_seq::body();
62 `uvm_info(get_type_name(), "Default sequence starting", UVM_HIGH)
63
64 repeat(10) begin
65 req = ahb_seq_item::type_id::create("req");
66 start_item(req);
67 if (!req.randomize() with {req.trans == NONSEQ; req.burst == INCR4;})
68 `uvm_error(get_type_name(), "Failed to randomize transaction")
69 finish_item(req);
70 end
71 req = ahb_seq_item::type_id::create("req");
72 start_item(req);
73 if (!req.randomize() with {req.trans == IDLE;})
74 `uvm_error(get_type_name(), "Failed to randomize transaction")
75 finish_item(req);
76 `uvm_info(get_type_name(), "Default sequence completed", UVM_HIGH)
77 endtask : body
switch success
verwcj@VM01:~/kelle...
teaching_sim
Makefile (~/keller_wor...
ahb_monitor.sv (~/kell...
[ahb_sramc_uvmtb_d...
ahb_driver.sv (~/kelle...
ahb_seq.sv + (~/kelle...
<Verdi:nTraceMain:1...